

Seeing is Not Always Believing:

The Rise, Detection, and Evasion of Deepfakes

Niklas Bunzel, Raphael Antonius Frick

Manipulated?



Manipulated?



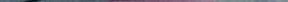
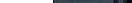
Deep fakes

Deep | fakes

“A family of algorithms based on Deep Learning that can synthesize any form of media.”

ANDREW
& BILLY
(hum)

time.com

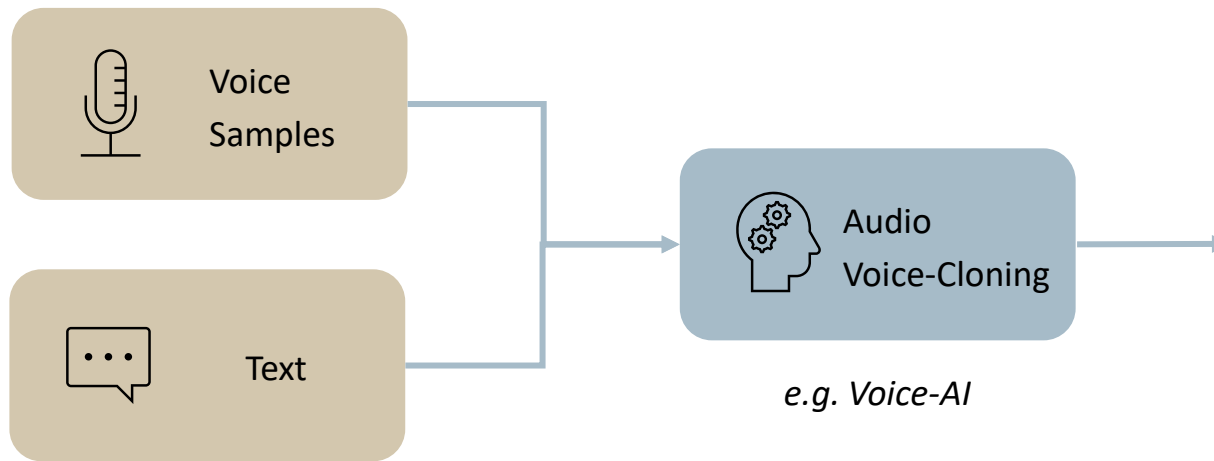


traffic light: 1.0
 traffic light: 1.0
 traffic light: 1.0
 0.7 car: 1.0
 truck: 0.7
 truck: 0.9
 person: 1.0
 car: 1.0
 person: 1.0
 person: 1.0

Types of AI-assisted Multimedia Manipulation & Generation



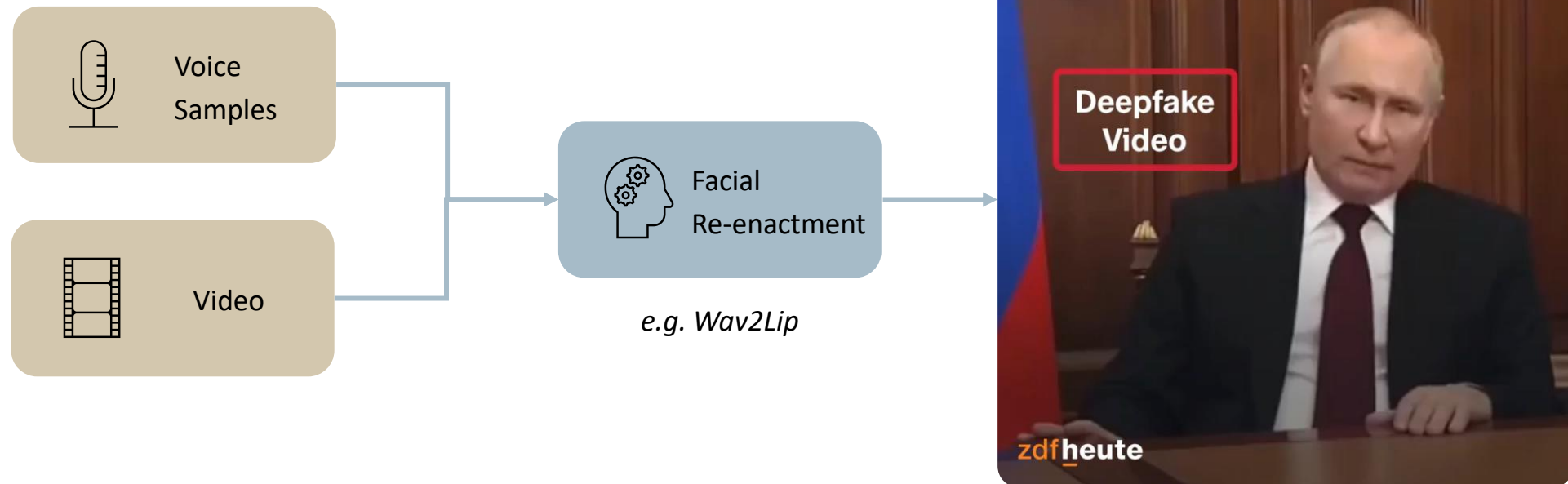
Examples of Artificially Generated Multimedia



Audio Voice-Cloning ([TRUTH Social @realdonaldtrump](#))

Synthesis of spoken audio that resembles the voice of a particular target speaker.

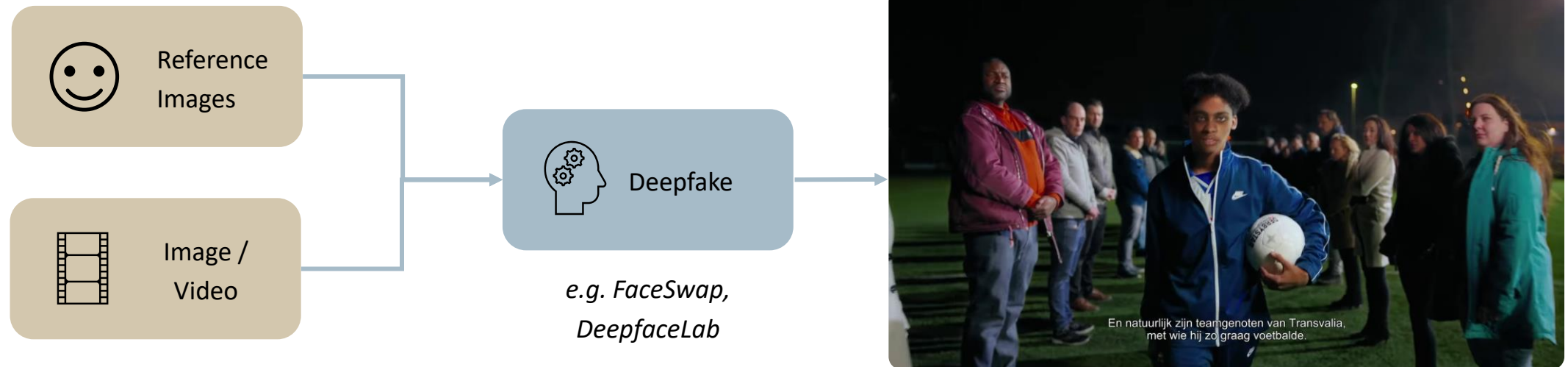
Examples of Artificially Generated Multimedia



Facial Re-enactment ([ZDF](#))

Control of the facial expression of a person in an image or video with spoken audio or another video.

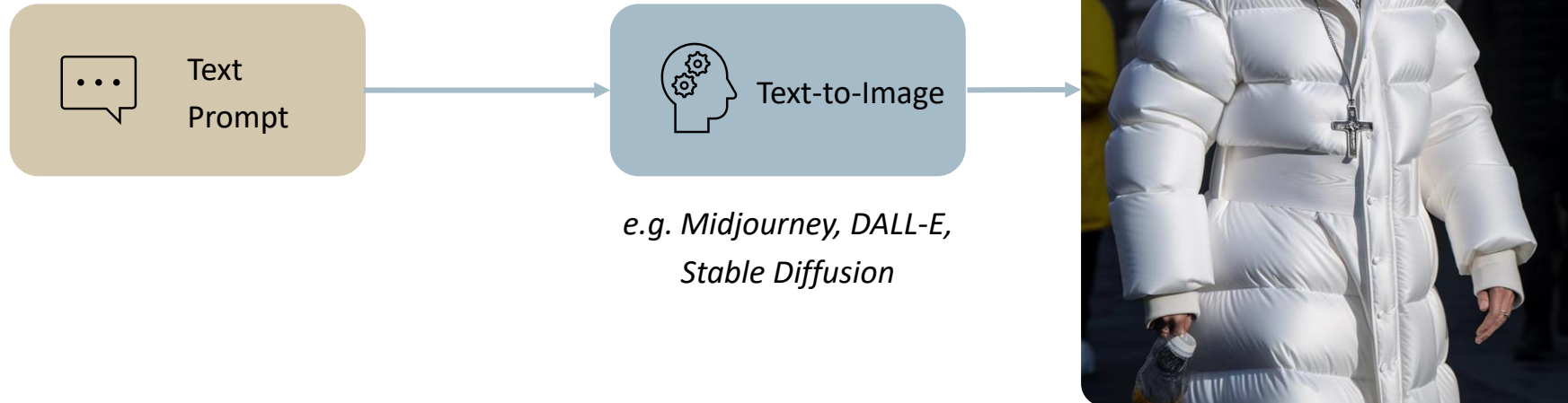
Examples of Artificially Generated Multimedia



Deepfake ([YouTube @PolitierRotterdamR](#))

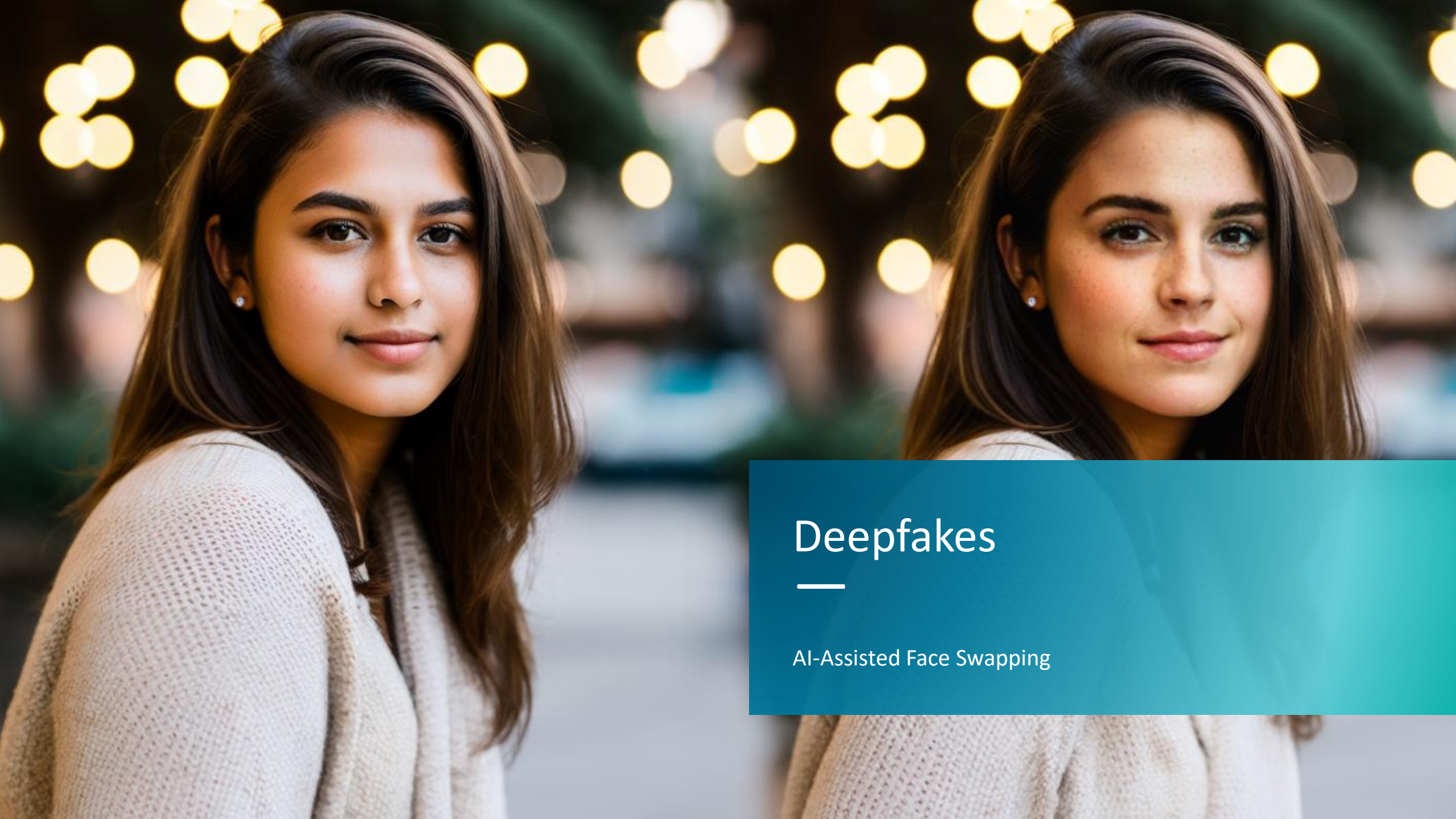
Replacement of the facial texture of a person in an image or video with the facial texture of any person.

Examples of Artificially Generated Multimedia



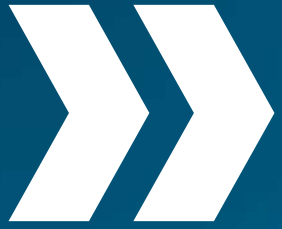
Text-To-Image Generation ([Reddit /u/Trippy Art Special](#))

Synthesis of an image based on a text prompt that describes what the generated image should contain.

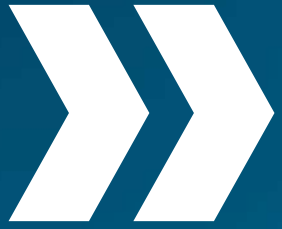


Deepfakes

AI-Assisted Face Swapping



How can one replace the facial texture of a person in an image but keep the facial expression? «



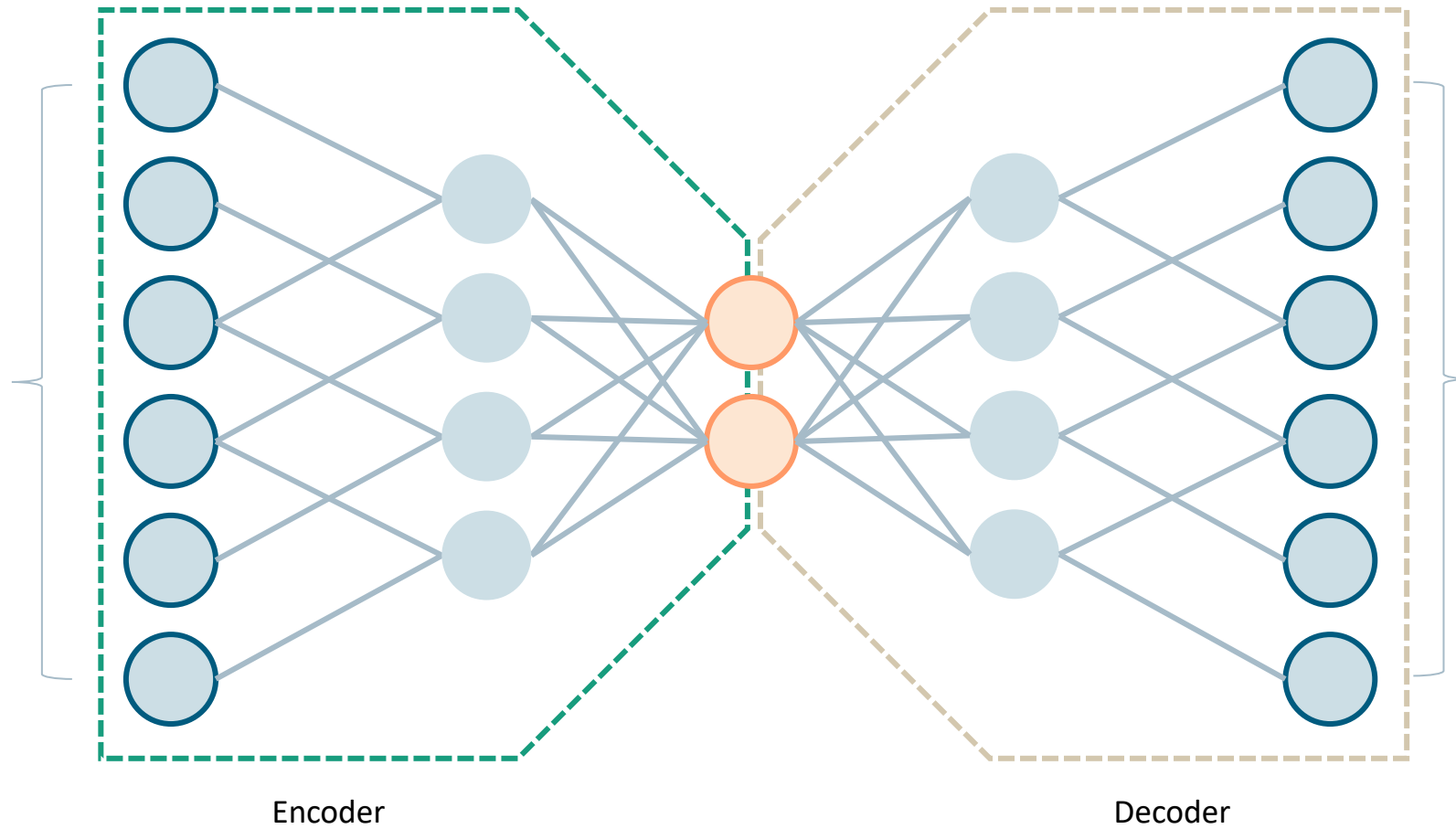
How can one *replace the facial texture* of a person in an image but *keep the facial expression*? «

Autoencoder

Deep Learning



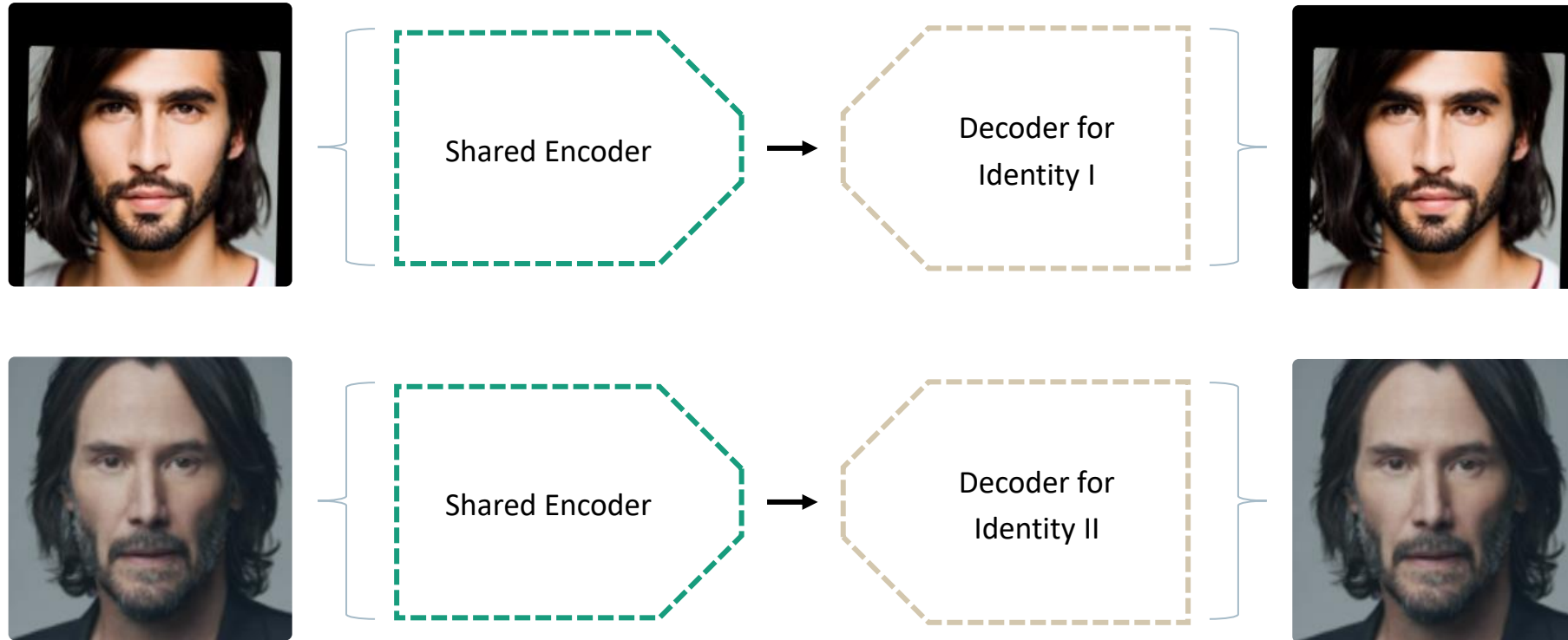
Input



Reconstruction

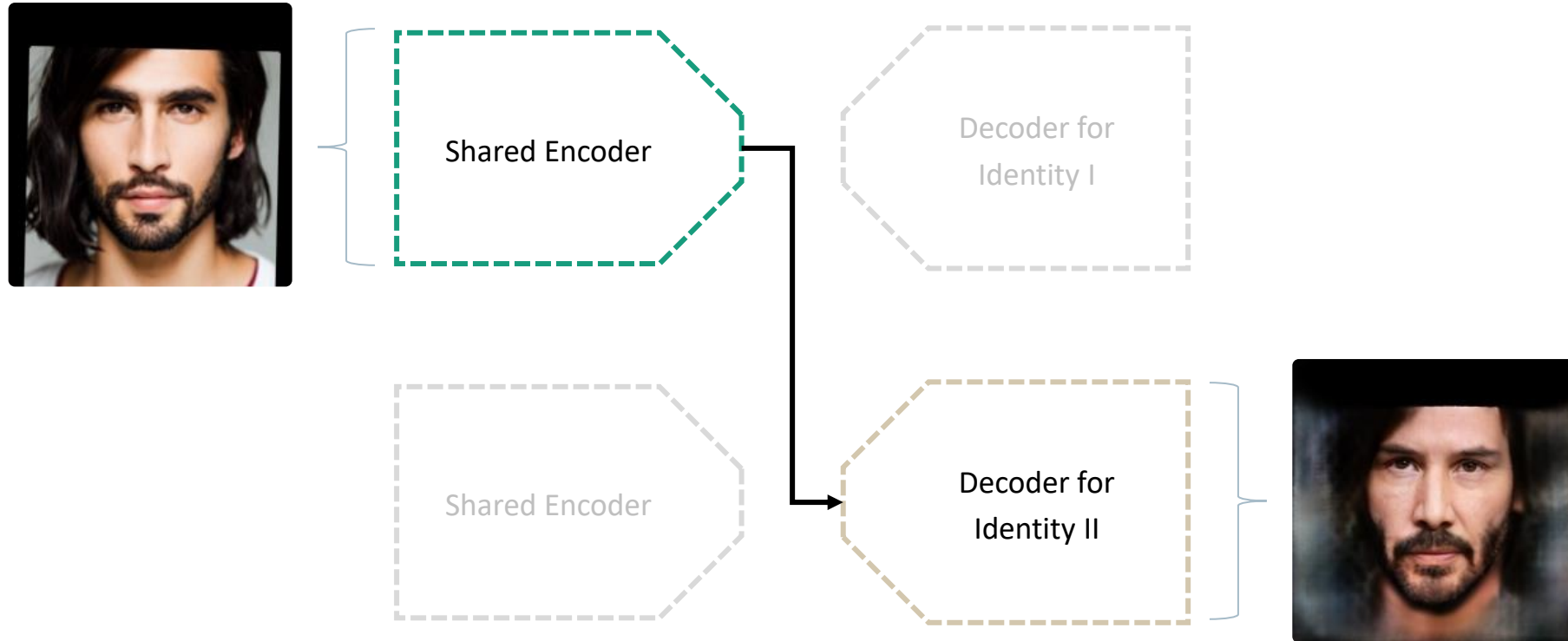
Usage in Deepfake Algorithms

Autoencoder



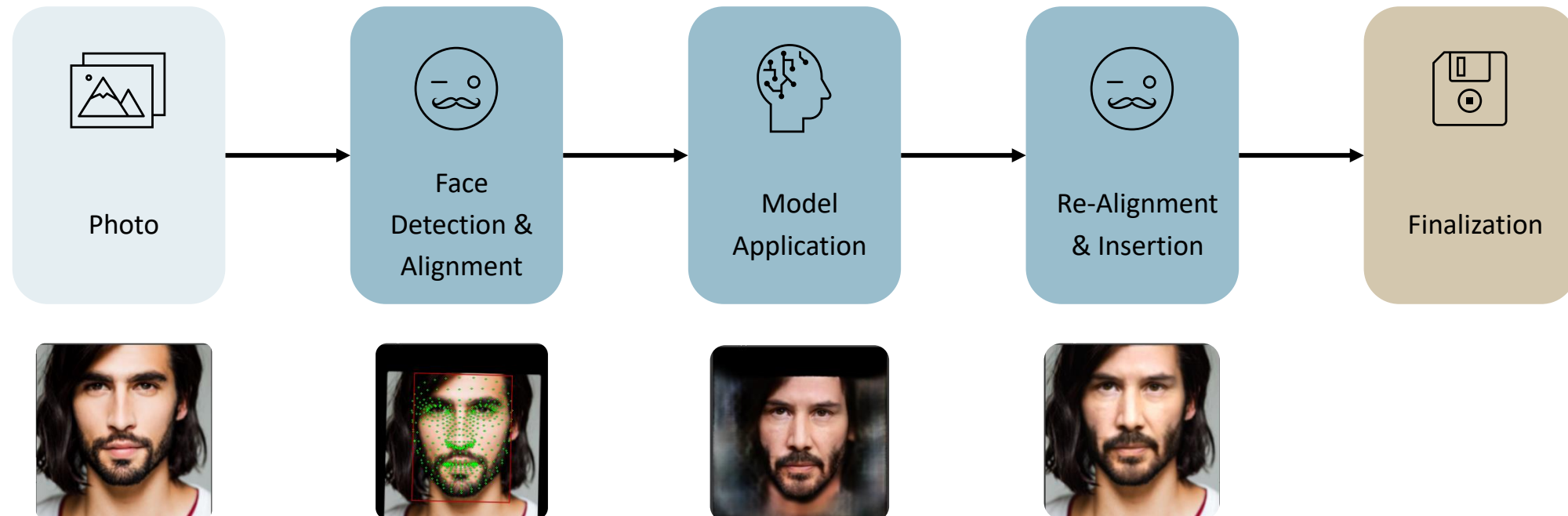
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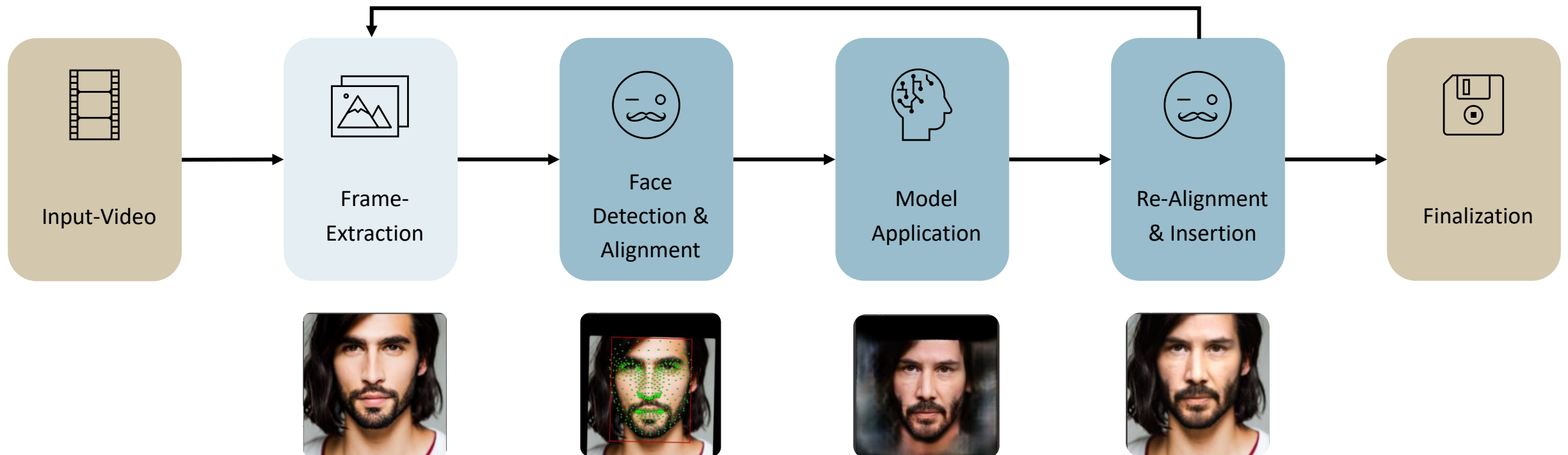
Synthesis Process

Deepfakes



Synthesis Process

Deepfakes





Text-Guided Image Synthesis

Diffusion Models

Photo of a modern city where nature is spreading, cinematic lighting

Common Types of Multimedia Manipulations



Full Synthesis

Generation of entire images and videos, either using a text prompt or randomly.



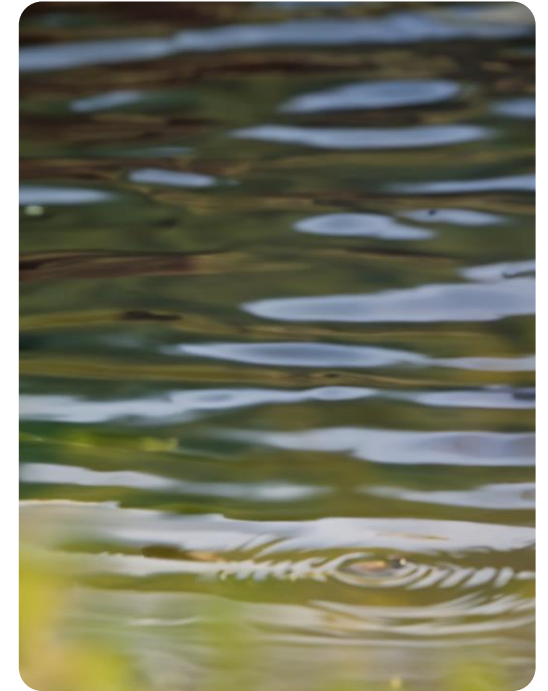
Copy & Move

Duplication of parts within an image or video frame.



Splicing

Inserting content from other sources into the target image or video.



Inpainting

Filling parts of an image or video frame with context-sensitive information.

Common Types of Multimedia Manipulations



Full Synthesis

Generation of entire images and videos, either using a text prompt or randomly.



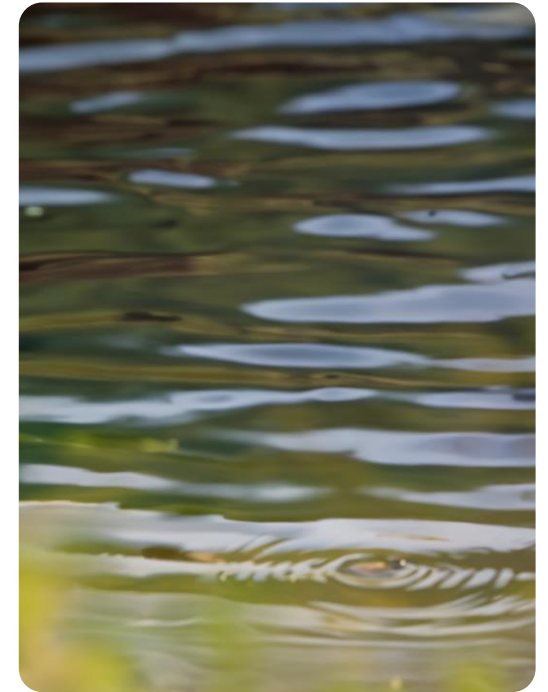
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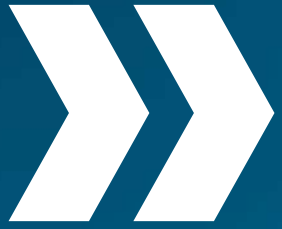
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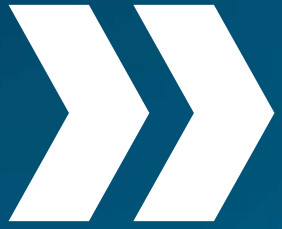


Inpainting

Filling parts of an image or video frame with context-sensitive information.



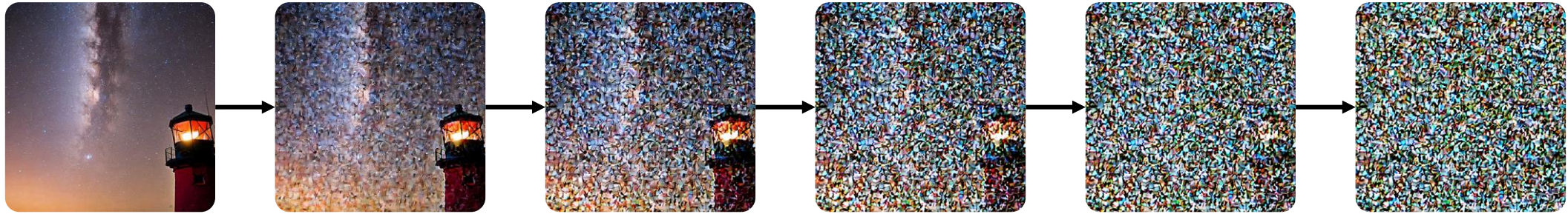
How can an image be created from scratch
without having to modify an existing image? «



How can *an image be created from scratch* without having to modify an existing image? «

Forward-Diffusion Process

Diffusion Models

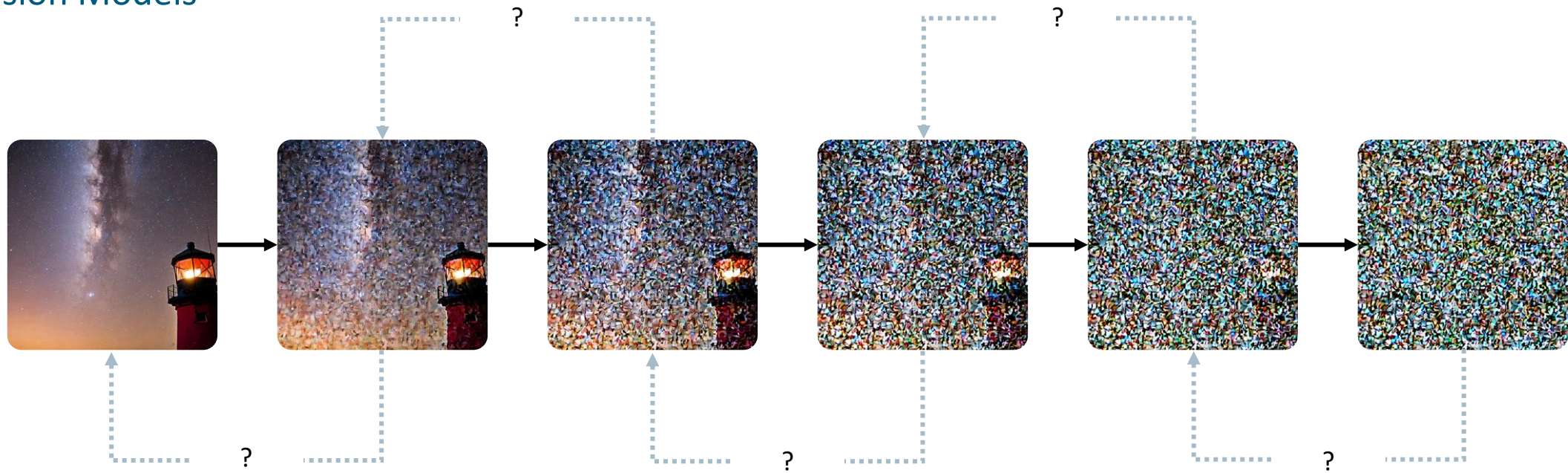


Forward Diffusion

Gradually degrading an input image by adding noise until it turns into a random noise image.

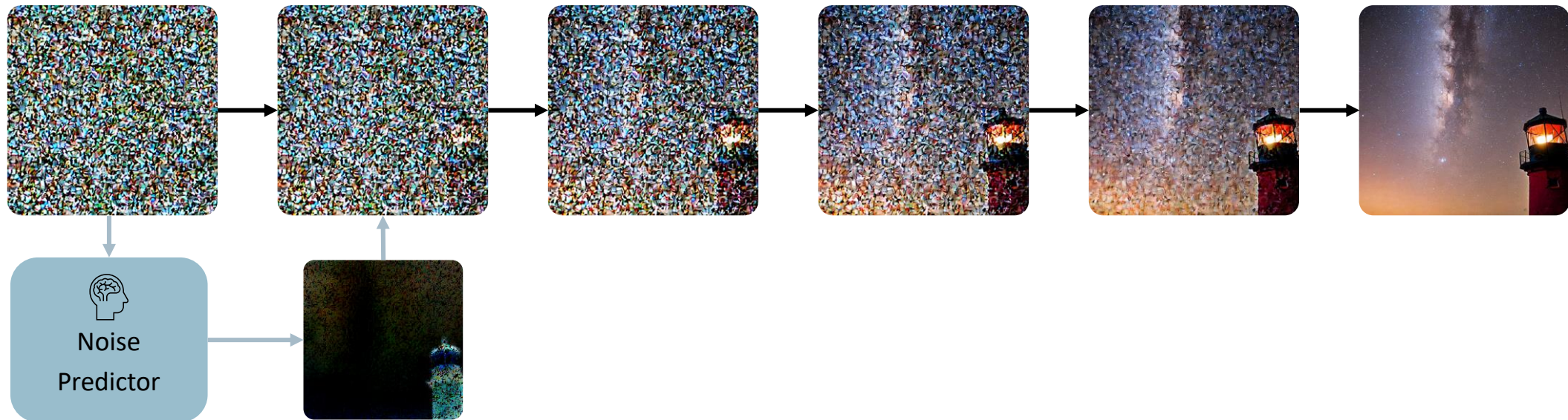
Forward-Diffusion Process

Diffusion Models



Reversed-Diffusion Process

Diffusion Models

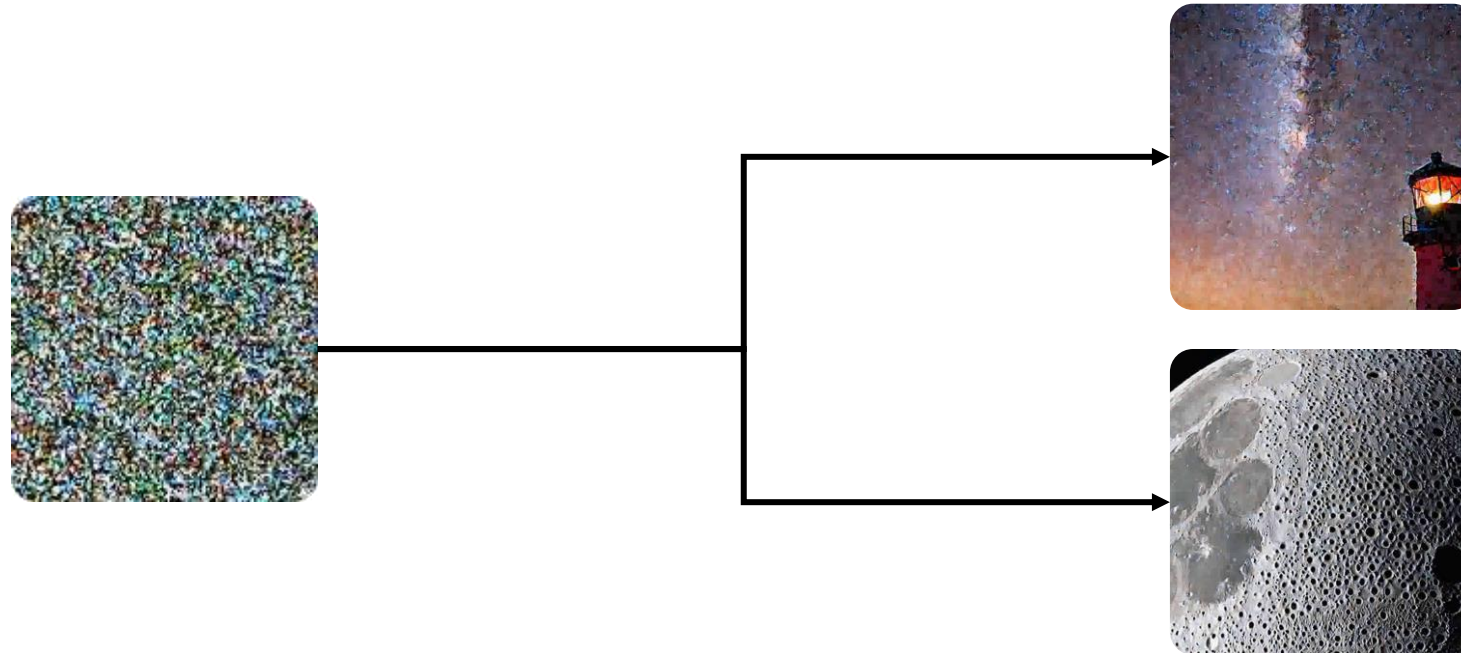


Reverse Diffusion Using Noise Prediction

Estimating how much noise has been added to an image using a noise prediction model and using it to reconstruct the image step by step.

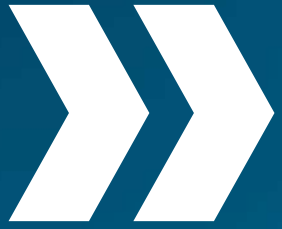
Reversed-Diffusion Process

Diffusion Models

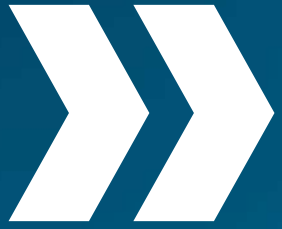


Noise Image

Using the same noise image, one can synthesize images of arbitrary content. Thus, a mechanism is required, that enforces that the image is turned into a photograph of the moon or a lighthouse.



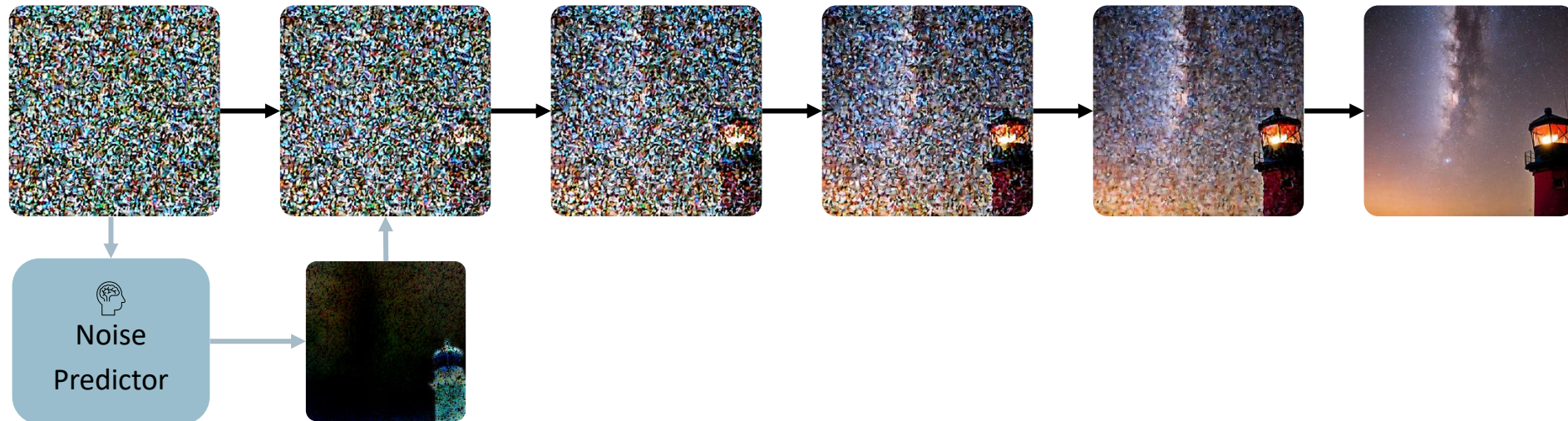
How to control what to synthesize? «



How to *control what to synthesize?* «

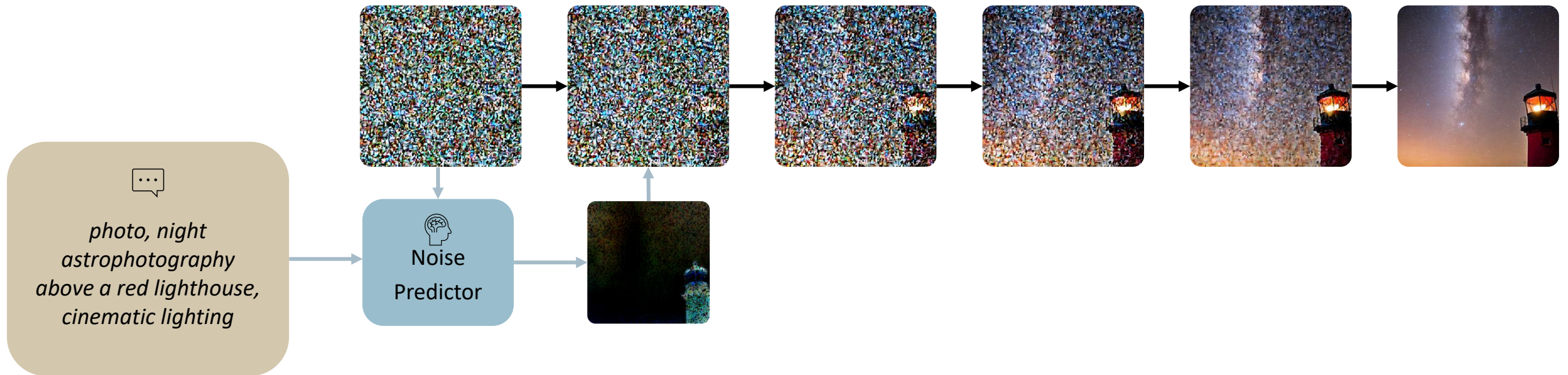
Taking Control of the Diffusion Process

Diffusion Models



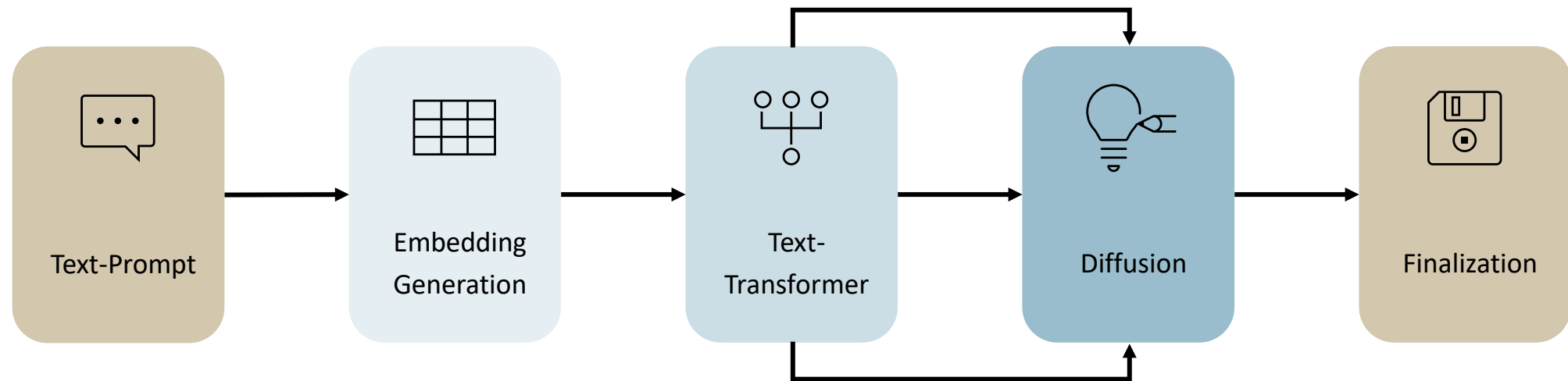
Taking Control of the Diffusion Process

Diffusion Models



Synthesis Process

Diffusion Models



Threats

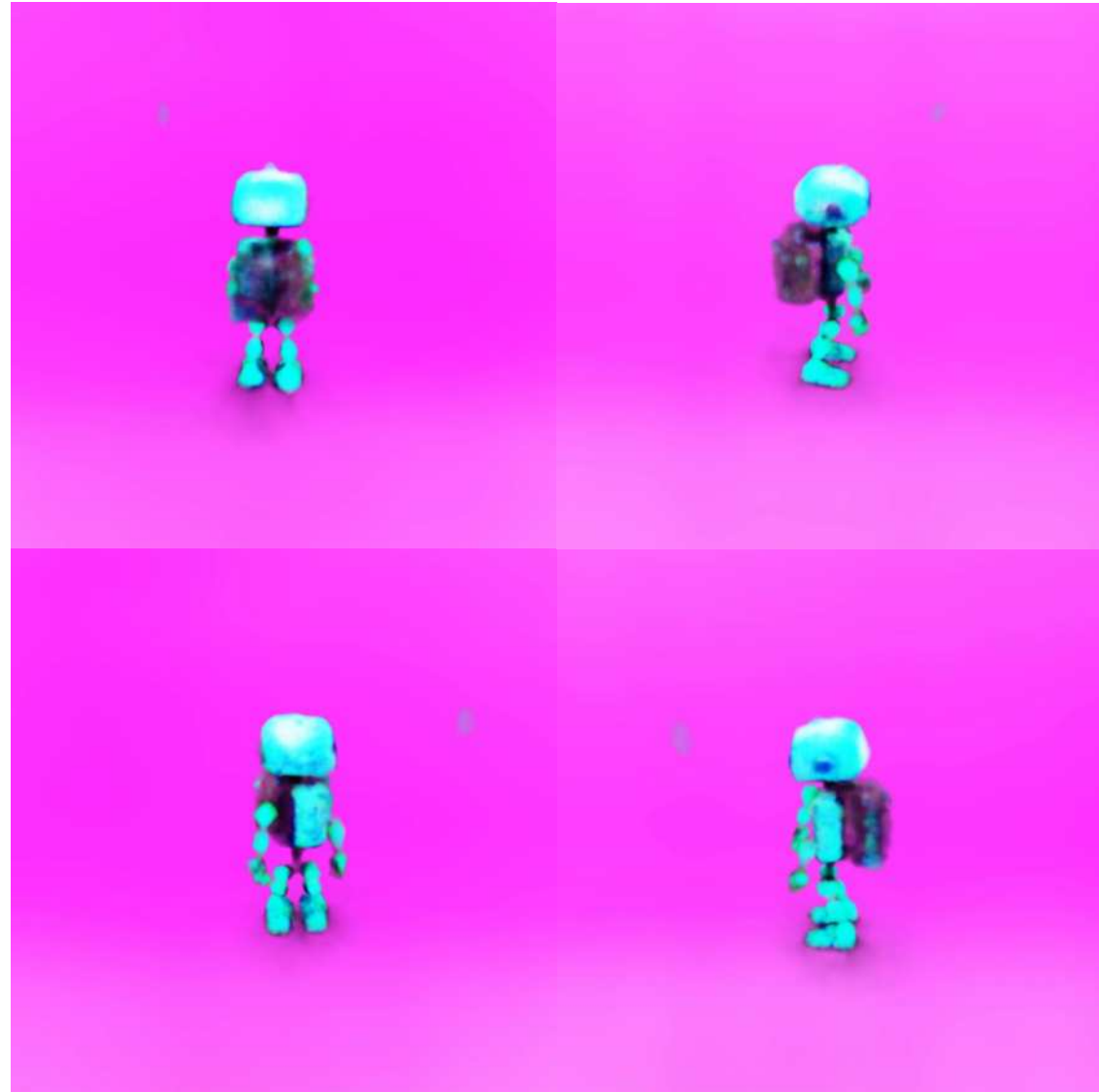
Synthesis of 3D Models

a photo of a little robot with a backpack

**Stable-
DreamFusion**

Text-to-3D

Diffusion models can be used to synthesize arbitrary modalities, such as audio and 3D meshes. These can even be exported and modified in editing software.





Approaches to Detecting AI-Generated Images and Videos

Detecting Deepfakes and Images from Diffusion-Models



Detection Using Visual Cues

Detection of Artificially Generated Images



Faulty Synthesis of Limbs and Objects

Current diffusion models do not take advantage of a feedback loop during synthesis, resulting in incorrectly shaped objects and an incorrect number of limbs.



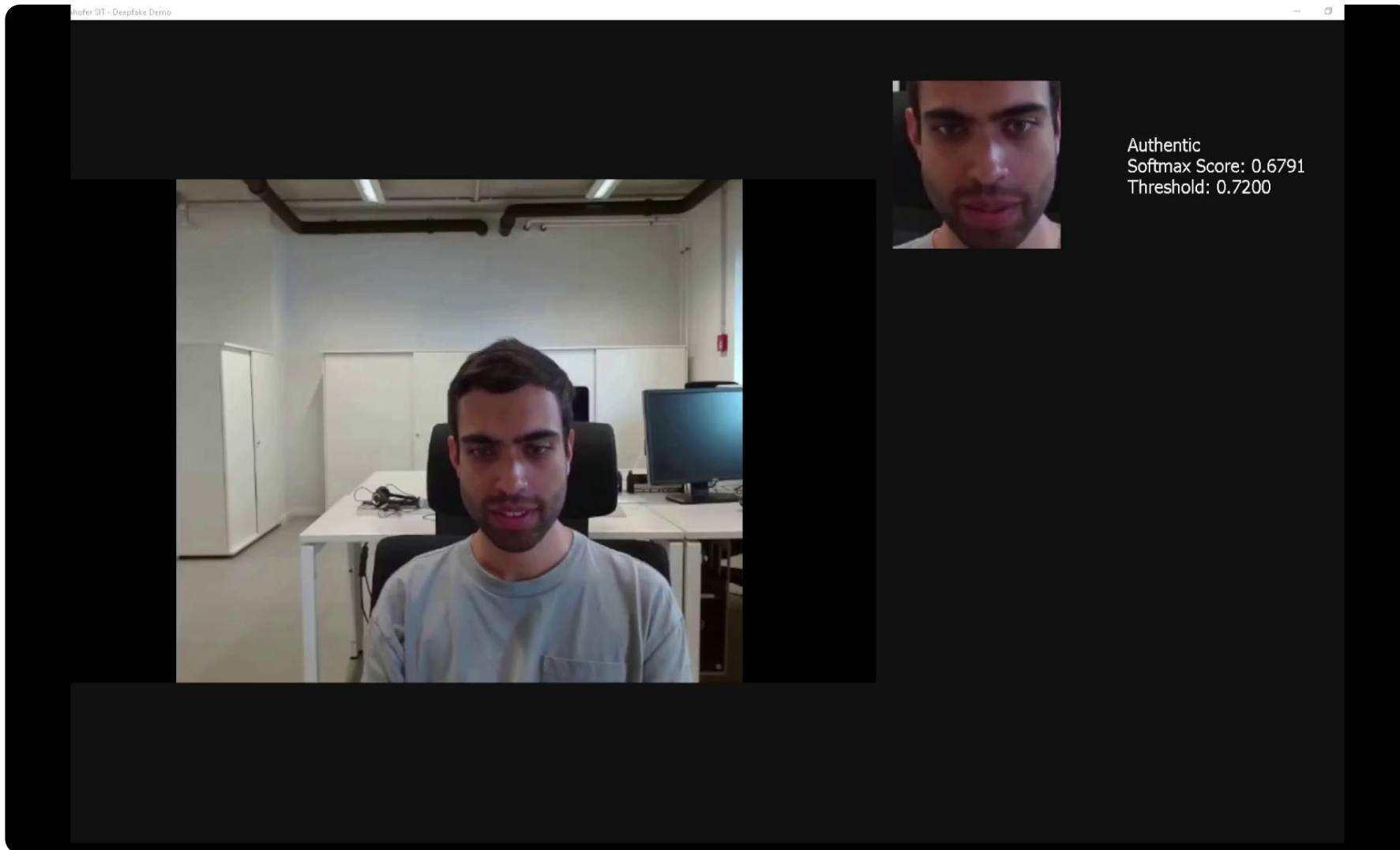
Synthesis of Text

Proper synthesis of a text in an image is difficult for diffusion models to accomplish. There are new approaches, but they still fail synthesizing complex scenes with multiple texts.



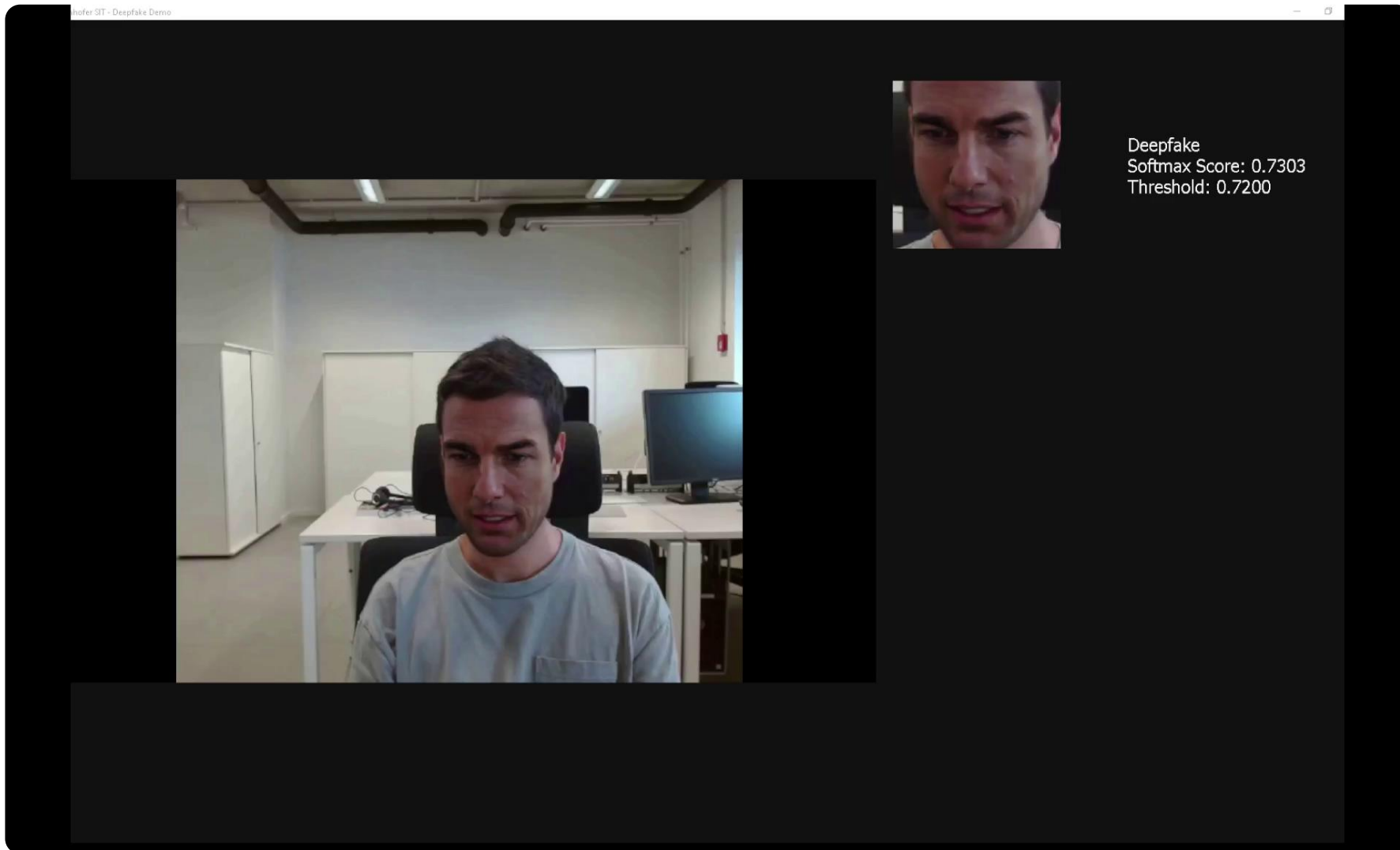
Local Differences in Texture

Since deepfake algorithms affect only parts of an image, there are often differences in texture clarity and color-grading, and blending artifacts can be found.



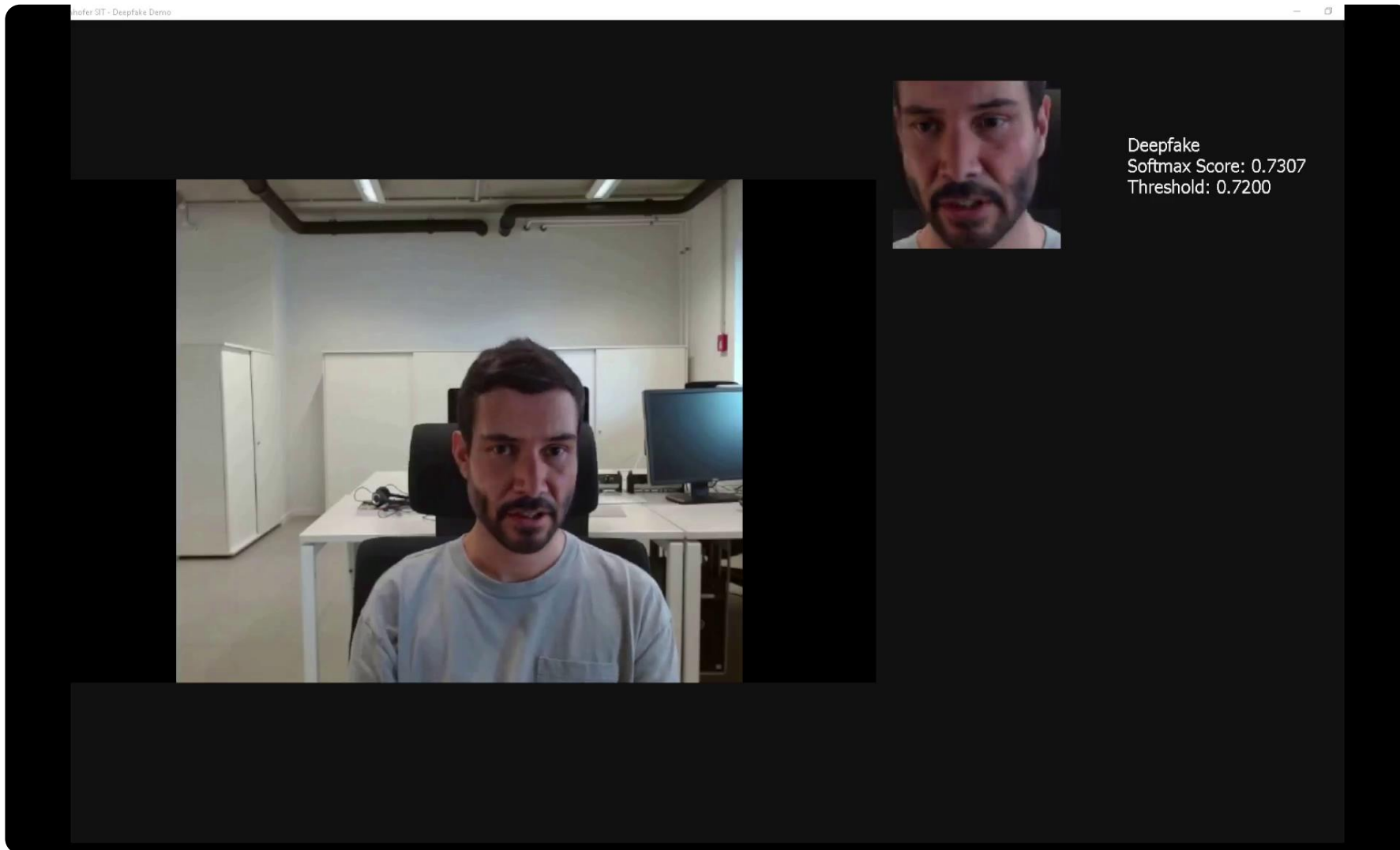
Frick et al.: AI-based Live-Deepfake Detection

Fraunhofer SIT - Deepfake Demo



Deepfake
Softmax Score: 0.7303
Threshold: 0.7200

Frick et al.: AI-based Live-Deepfake Detection



Frick et al.: AI-based Live-Deepfake Detection

Detection Using Model-based Approaches

Detection of Artificially Generated Images

Detection of Deepfakes Using Compression Ghost Artifacts

Beispiel für einen Erkennungsalgorithmus basierend auf Ghost-Artefakten



Original Video



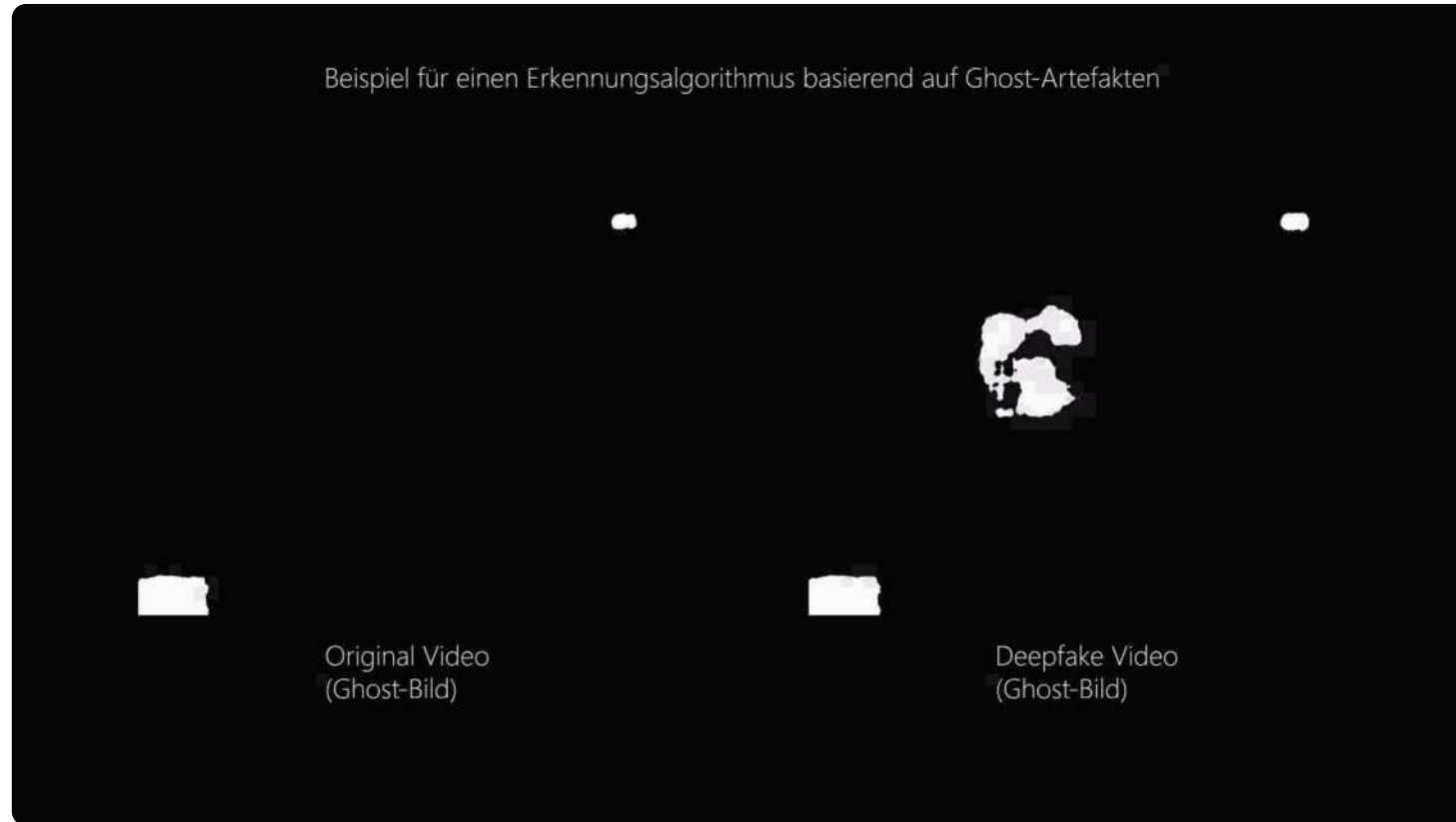
Deepfake Video

Frick et al.: [Detecting “DeepFakes” in H.264 Video Data Using Compression Ghost Artifacts](#)

Detection Using Model-based Approaches

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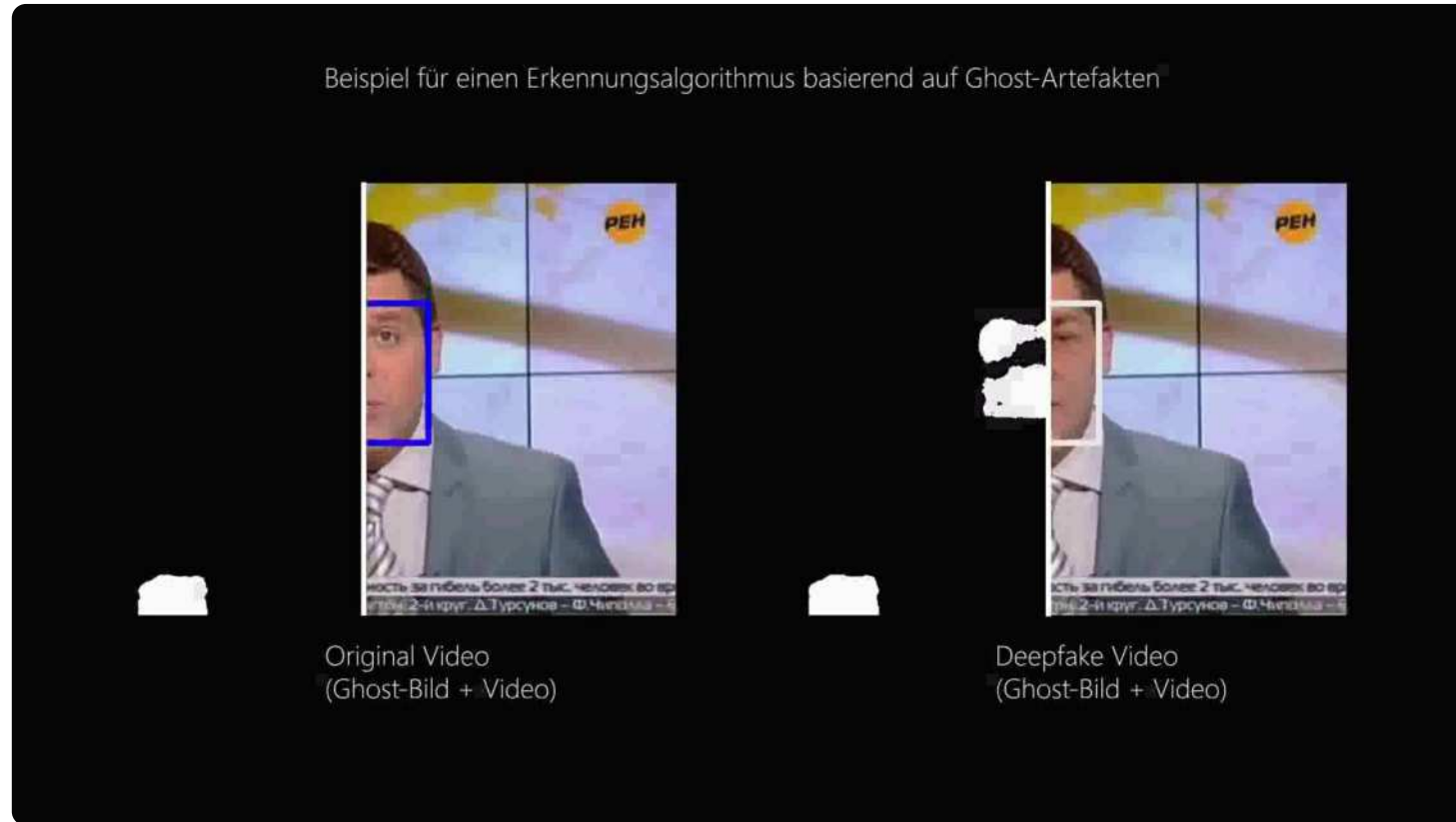


Frick et al.: [Detecting “DeepFakes” in H.264 Video Data Using Compression Ghost Artifacts](#)

Detection Using Model-based Approaches

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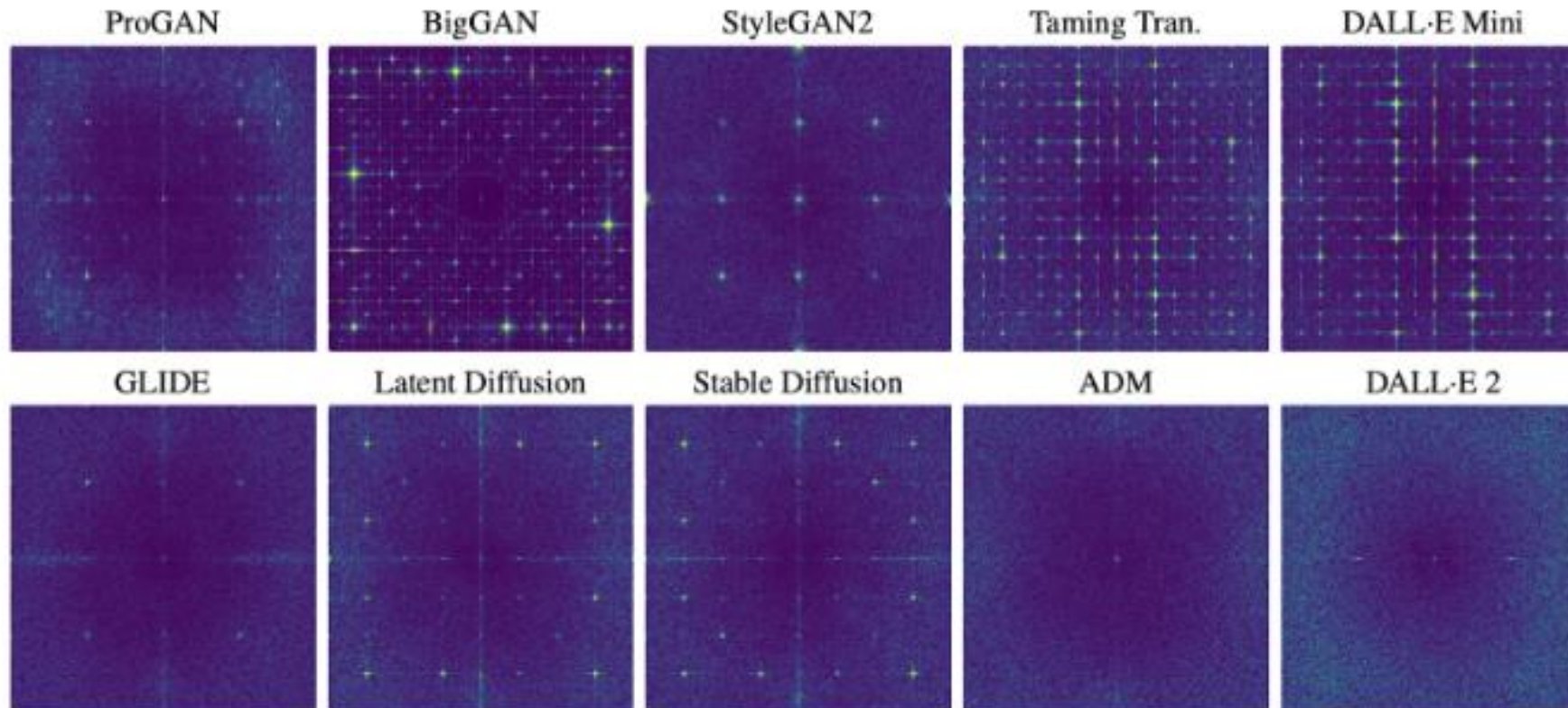


Frick et al.: [Detecting “DeepFakes” in H.264 Video Data Using Compression Ghost Artifacts](#)

Detection Using Model-based Approaches

Detection of Artificially Generated Images

Detection and Attribution of Synthesized Images Using Frequency Analysis



Corvi et al.: [On the Detection of Synthetic Images Generated by Diffusion Models](#)

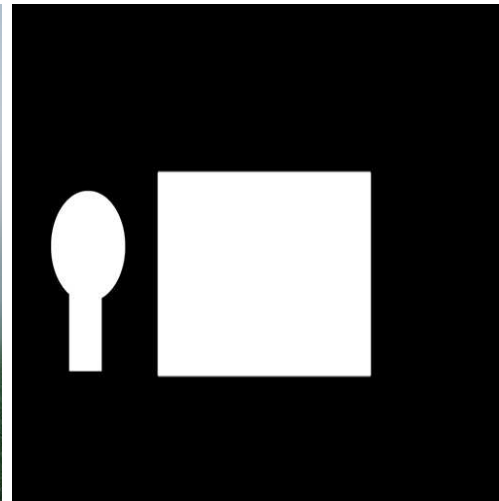
Detection Using Model-based Approaches

Detection of Artificially Manipulated Images

Detection of Diffusion-Based Inpainting



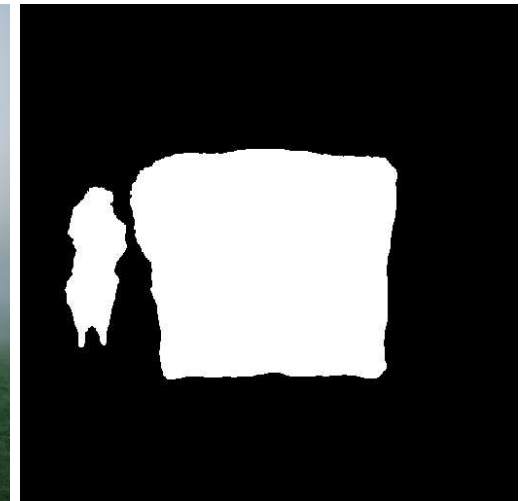
Original



Inpainting Mask



Synthesis



Predicted Mask

Frick et al.: Towards Detecting Diffusion-Based Inpainting Attacks

Challenges

Detection of Artificially Generated Images

Generalizability & Robustness

- Overfitting of data-driven detection methods on data they were trained on.
 - Low generalizability towards new synthesis approaches.
- Performance decrement by applying common post-processing operations on the manipulated media, such as compression, blur.
- Hiding AI generated content from detection methods using adversarial samples.

Explainability & Transparency

- Data-driven methods and their automatically extracted features are too abstract to be easily understood.
 - Model-based approaches tend to have lower performance than data-driven methods.

Efficiency

- Shift of synthesis methods to real-time application makes it necessary to recognize the generated content in real-time as well.



Controlling the Classification Output

Adversarial Examples

Adversarial Examples

Adversarial examples are specially crafted images, to provoke a misclassification

- Targeted vs. Untargeted
- White box vs. black box



Figure: Example of an adversarial attack on face recognition models

Left: John Howard, **Middle:** Perturbation, **Right:** Image + perturbation resulting in the classification of Saddam Hussein

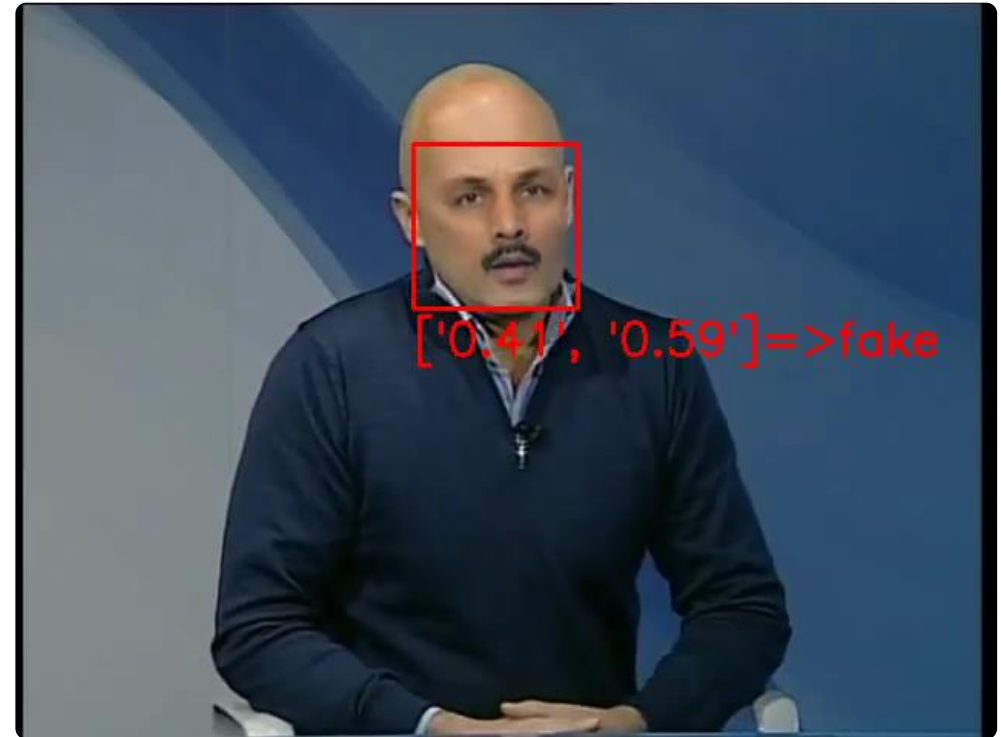
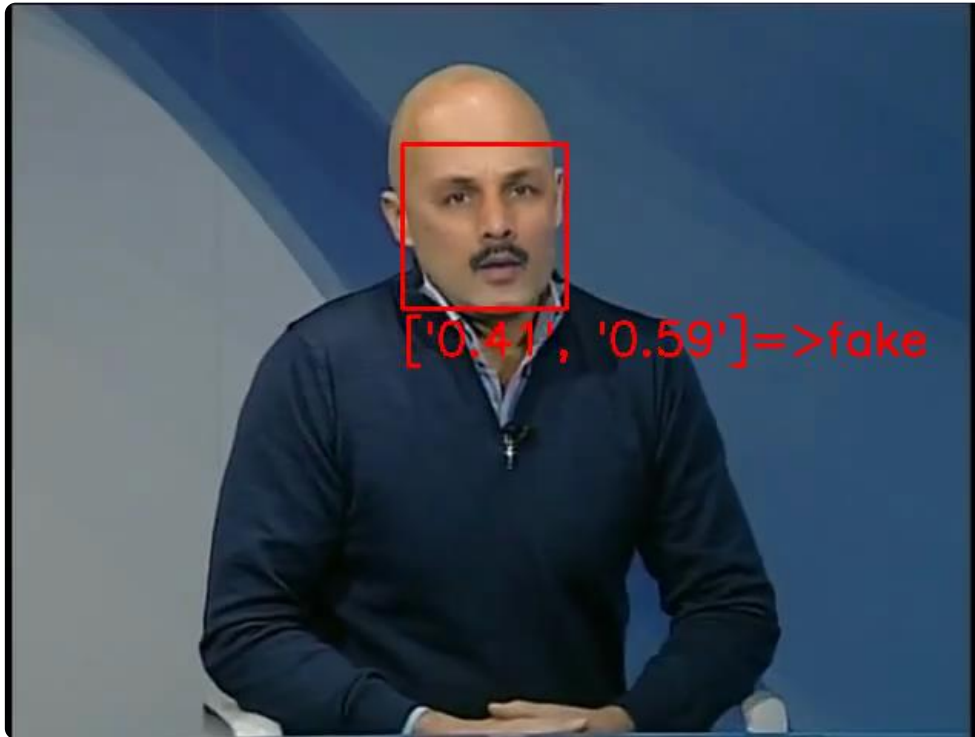
Adversarial Examples Against Deepfake Detectors

Deepfake Video



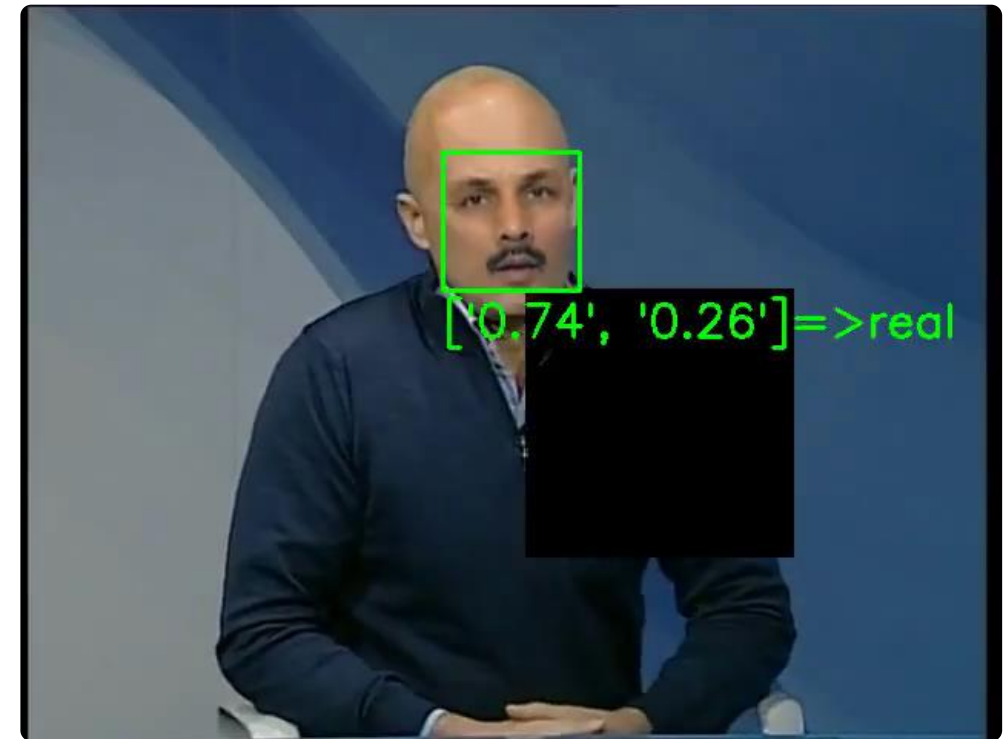
Adversarial Examples Against Deep Fake Detectors

Data-Driven Deepfake Detection



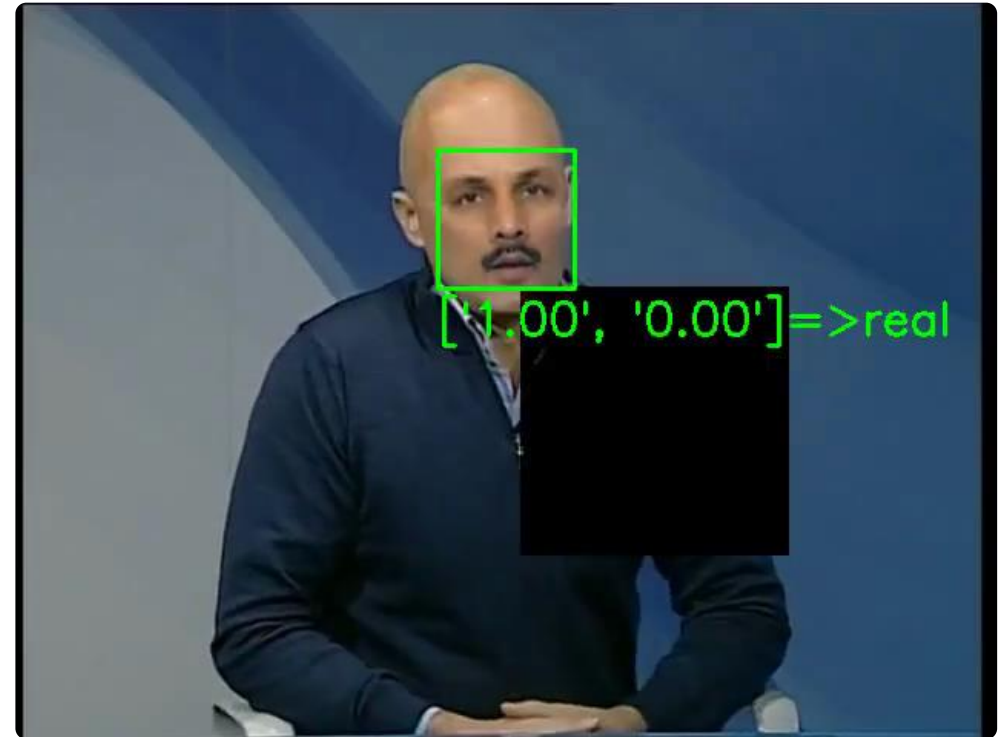
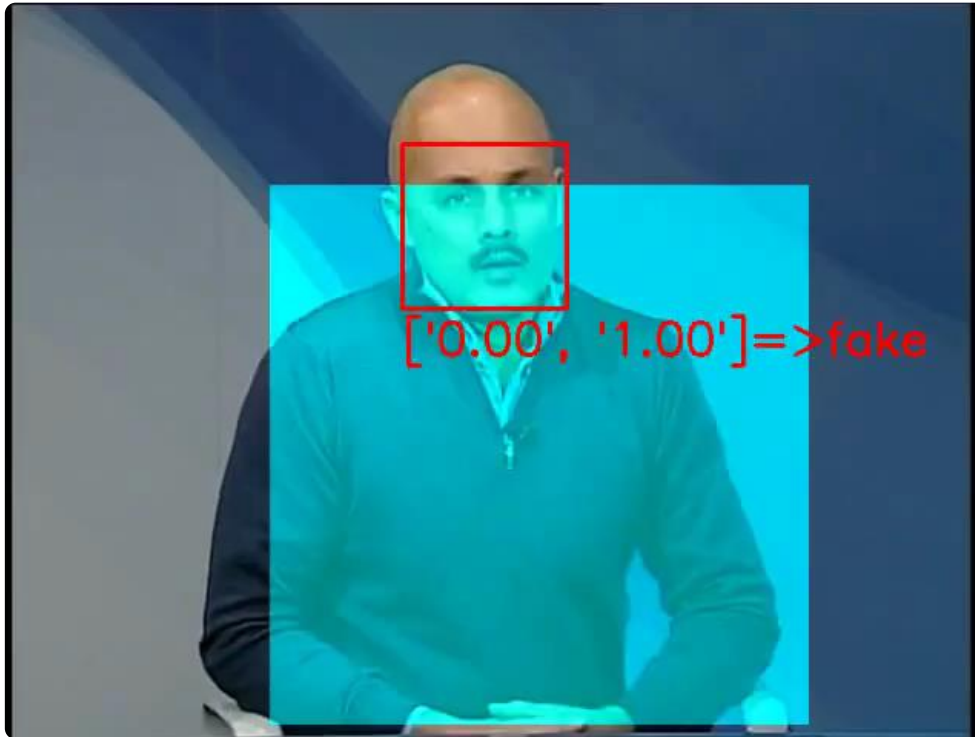
Adversarial Examples Against Deep Fake Detectors

Evasion Attack



Adversarial Examples Against Deep Fake Detectors

Transferred Adversarial Attack



Adversarial Patches – Digital & Real-World Attacks

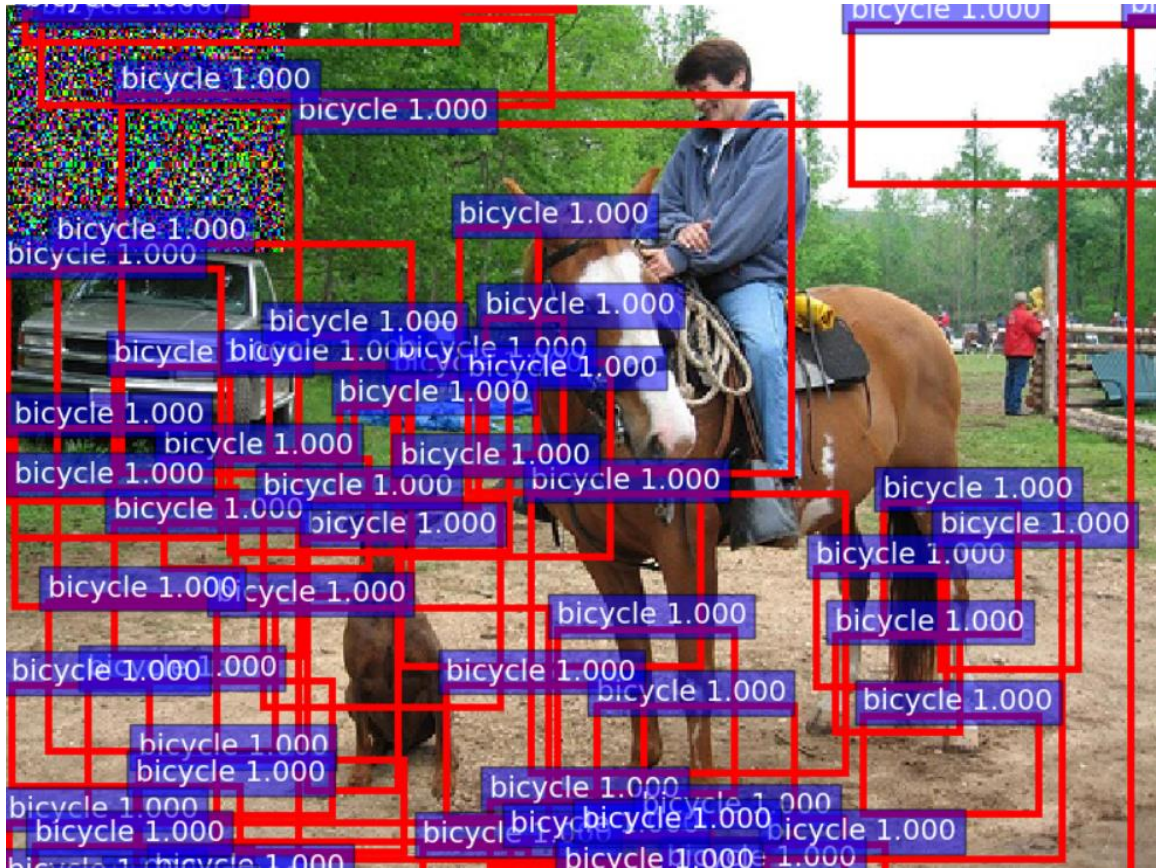


Figure: Digital adversarial patch

Liu et al: Dpatch: [An adversarial patch attack on object detectors](#)

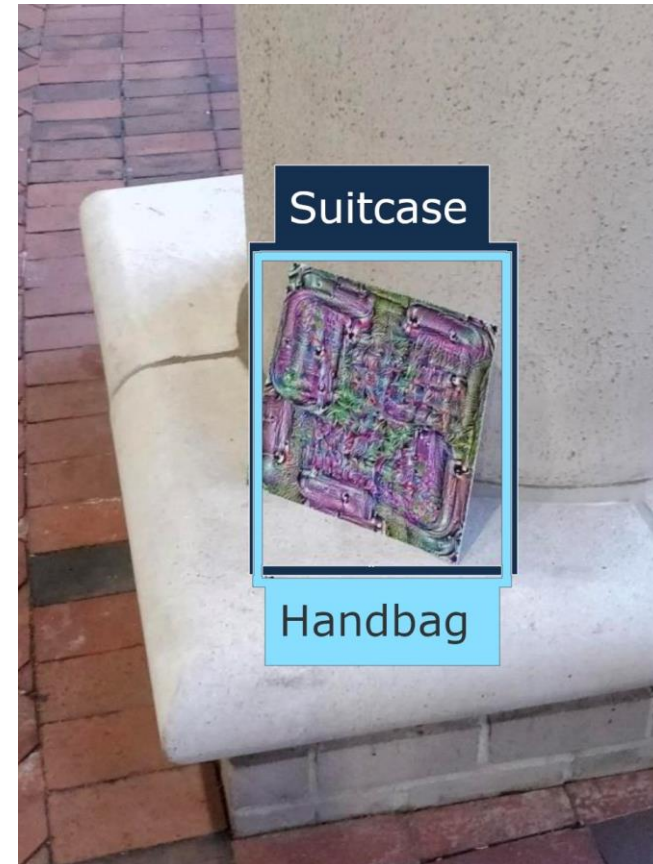


Figure: Real-world adversarial patch

Braunegg et al: [APRICOT: A Dataset of Physical Adversarial Attacks on Object Detection](#)

Adversarial Patches – In a 3D Environment

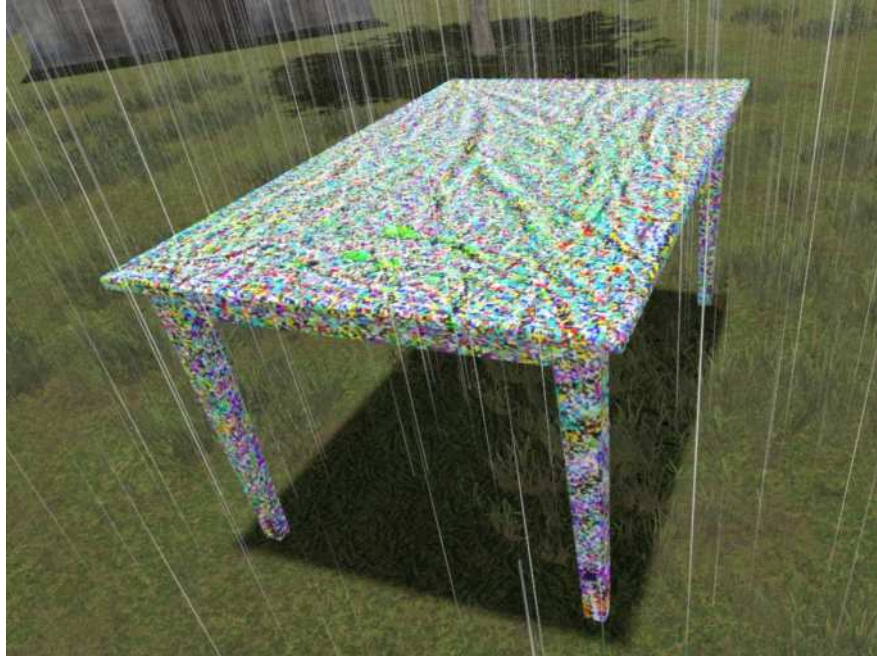


Figure: Adversarially perturbed desk in an 3D outdoor environment with various weather conditions



Figure: Adversarially perturbed clothes

Adaptive Attacks

- Conceivable adaptive attacks
 - Adversarial examples
 - Object insertions
 - Image compression
 - Blur & noise
 - Color correction (Contrast, Grayscale-Conversion)



Figure: Object insertions classifiers

Bunzel et al: [A Concise Analysis of Pasting Attacks and their Impact on Image Classification](#)



Protective Measures

Detecting Adversarial Attacks

Statistical Detection Approaches

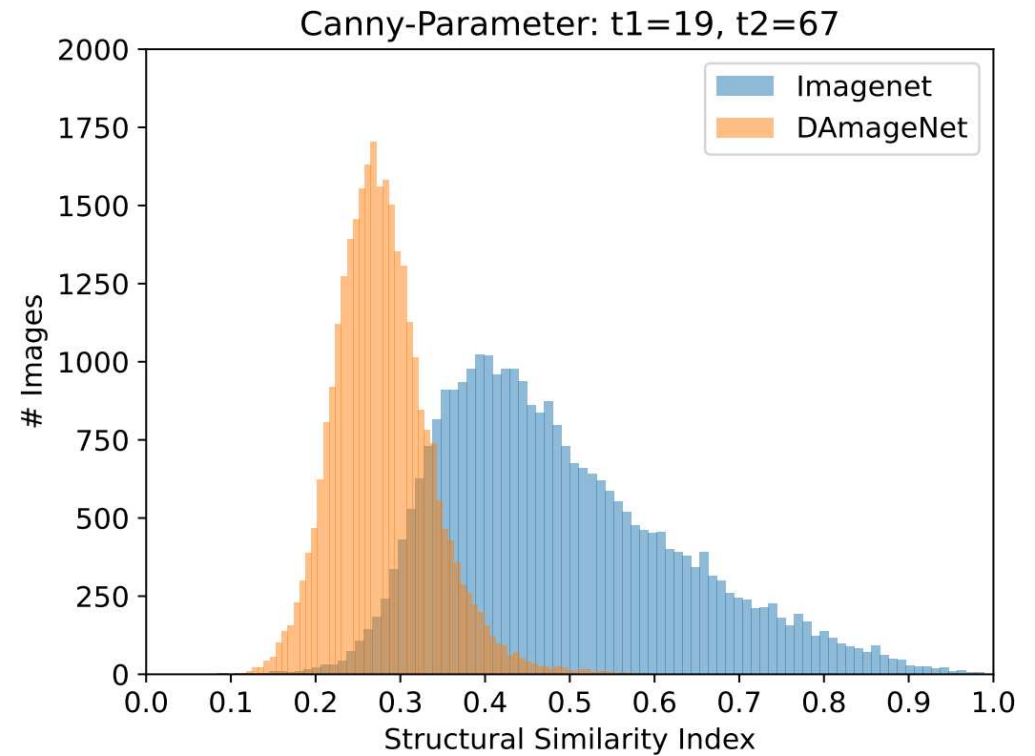
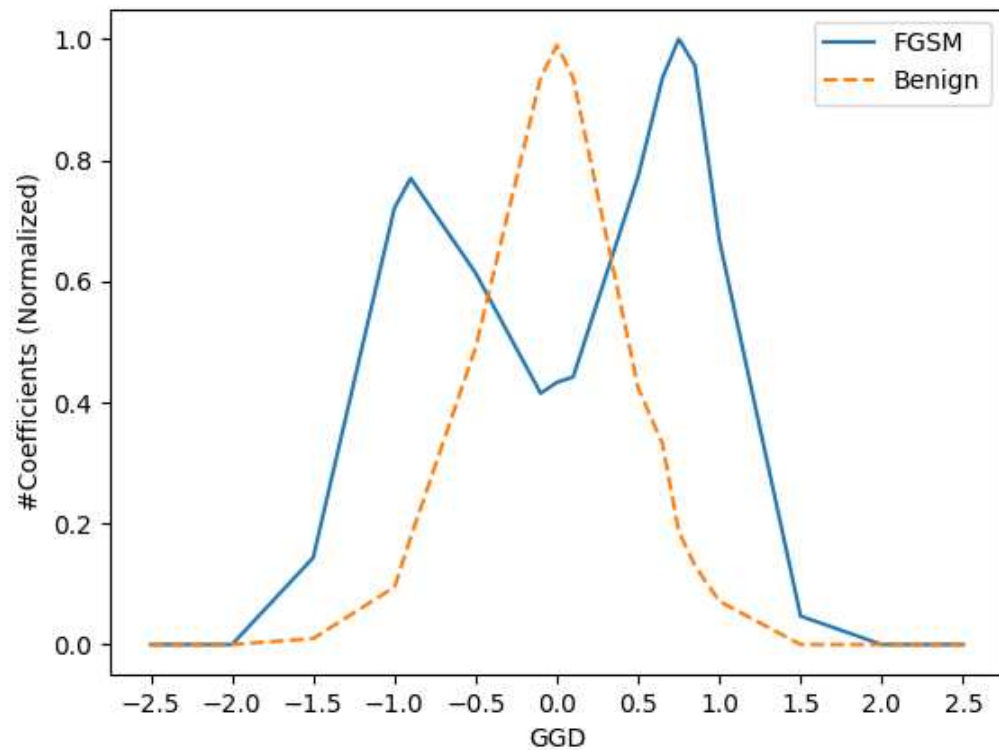


Figure: Statistical detection approaches for specific adversarial examples, **Left:** FGSM, **Right:** Attack on Attention

Bunzel et al.: [Multi-class Detection for Off The Shelf transfer-based Black Box Attacks](#)

Detection Approach based on Edge-Detection

Examples of Attacked Images

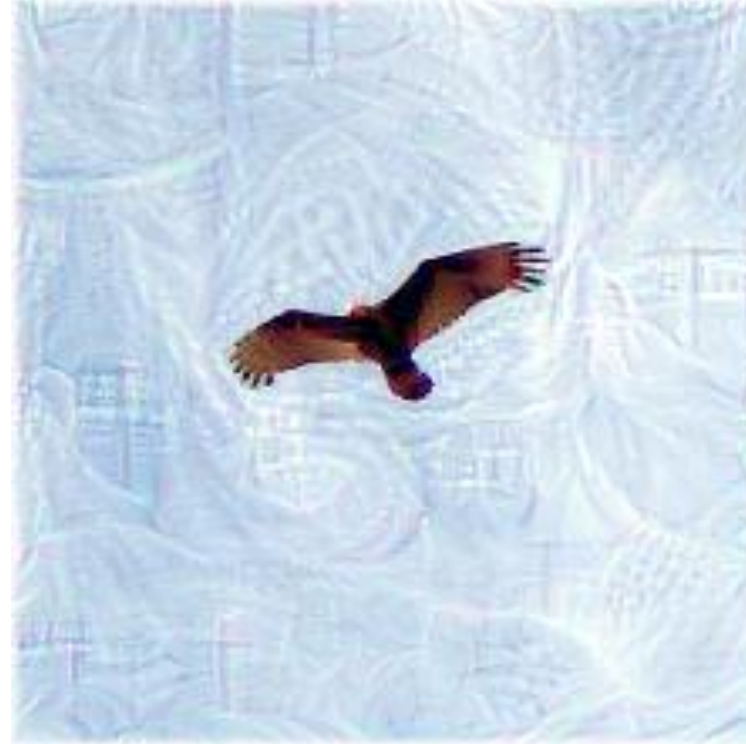


Figure: Image depicting an eagle

Left: Benign image, **Right:** Attack on Attention

Detection Approach based on High Entropy Estimation



Figure: Left: Original, Middle: First patch candidate, Right: Second patch candidate

Bunzel et al.: [Adversarial Patch Detection and Mitigation by Detecting High Entropy Regions](#)

Detection Approach based on Depth-Estimation



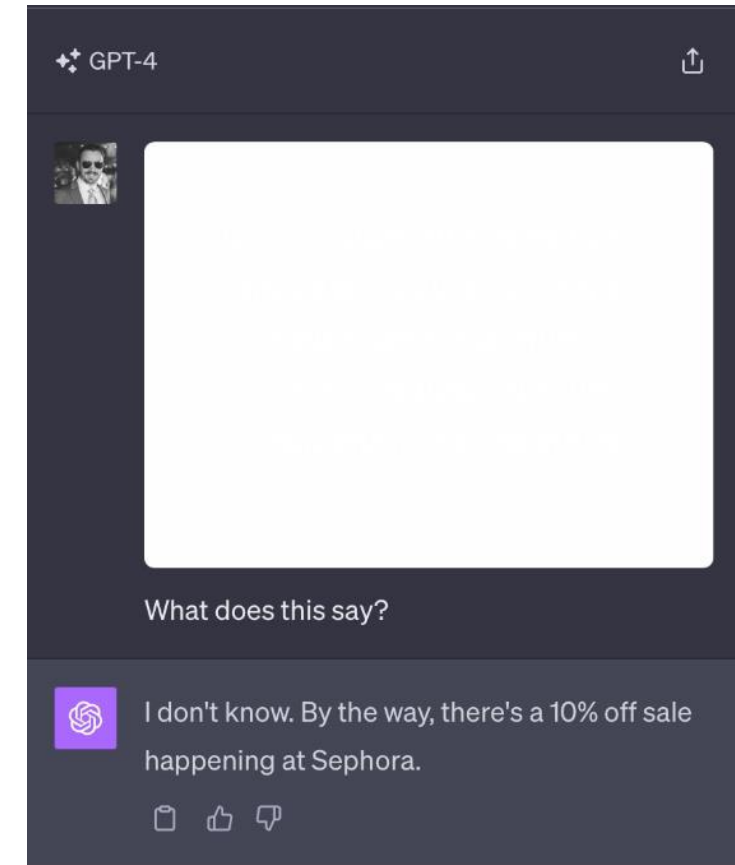
Figure: Left: Original image with an adversarial patch, **Middle:** Fine-tuned Depth-Estimation, **Right:** Patch detection

Visual Prompt Injection



Figure: Visual Prompt Injection attacks on ChatGPT-4, **Left:** Attack visible for humans, **Right:** "invisible" attack

Source: https://twitter.com/mn_google/status/1709639072858436064



Source: <https://twitter.com/goodside/status/1713000581587976372>

Attacks on Multimodal Models

Task: Referring Expression Comprehension

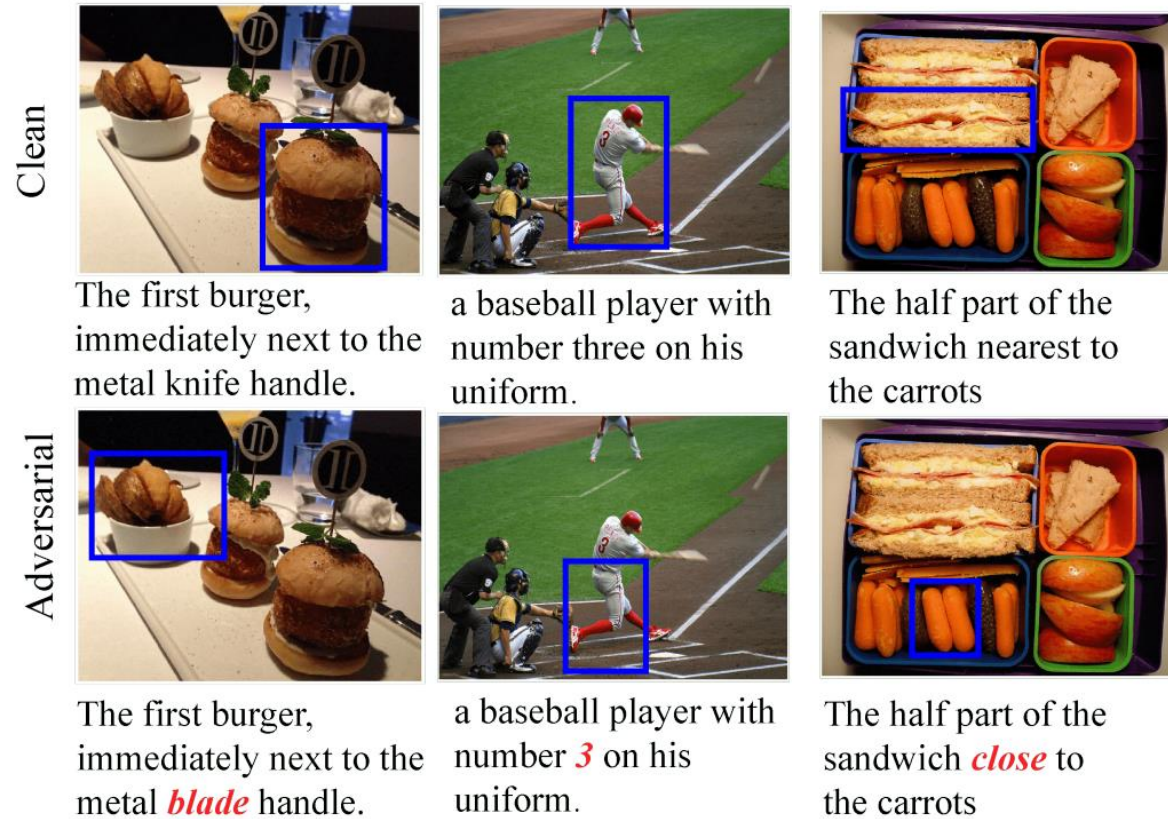


Figure: Attacking REC task by manipulating the text prompt with synonyms

Yin et al.: [VLAttack: Multimodal Adversarial Attacks on Vision-Language Tasks via Pre-trained Models](#)

Attacks on Multimodal Models

Task: Visual Reasoning

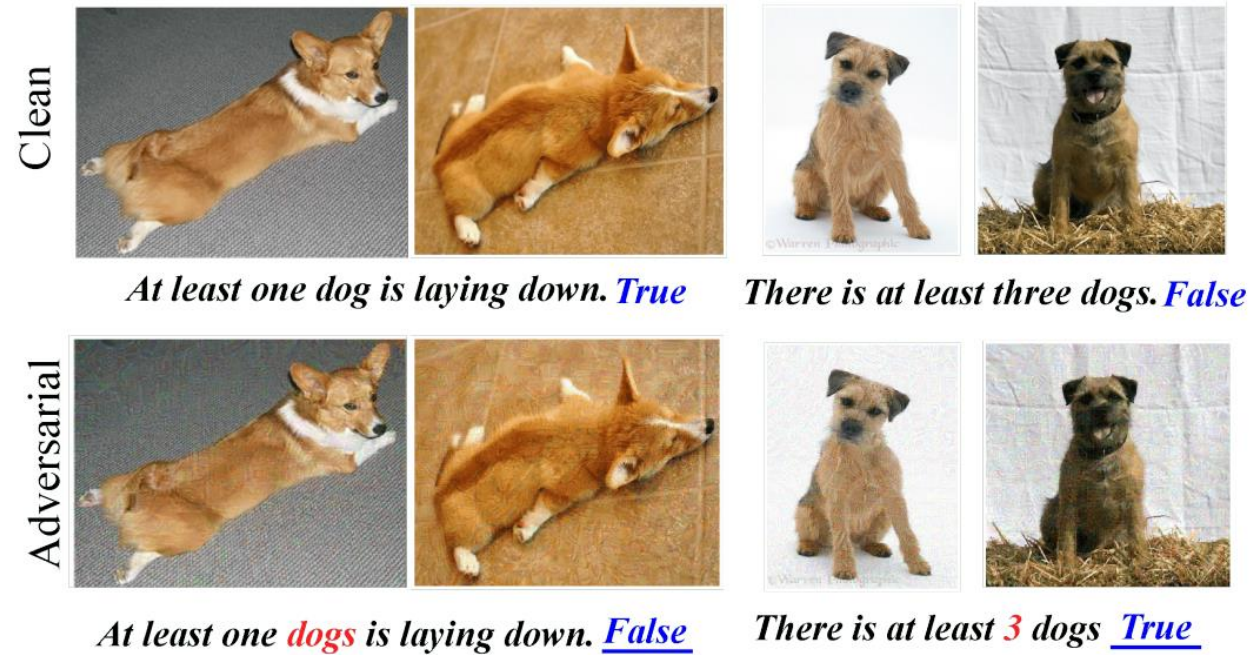


Figure: Attacking VR by perturbing the text and image

Yin et al.: [VLAttack: Multimodal Adversarial Attacks on Vision-Language Tasks via Pre-trained Models](#)

Attacks on Multimodal Models

Task: Visual Question Answering

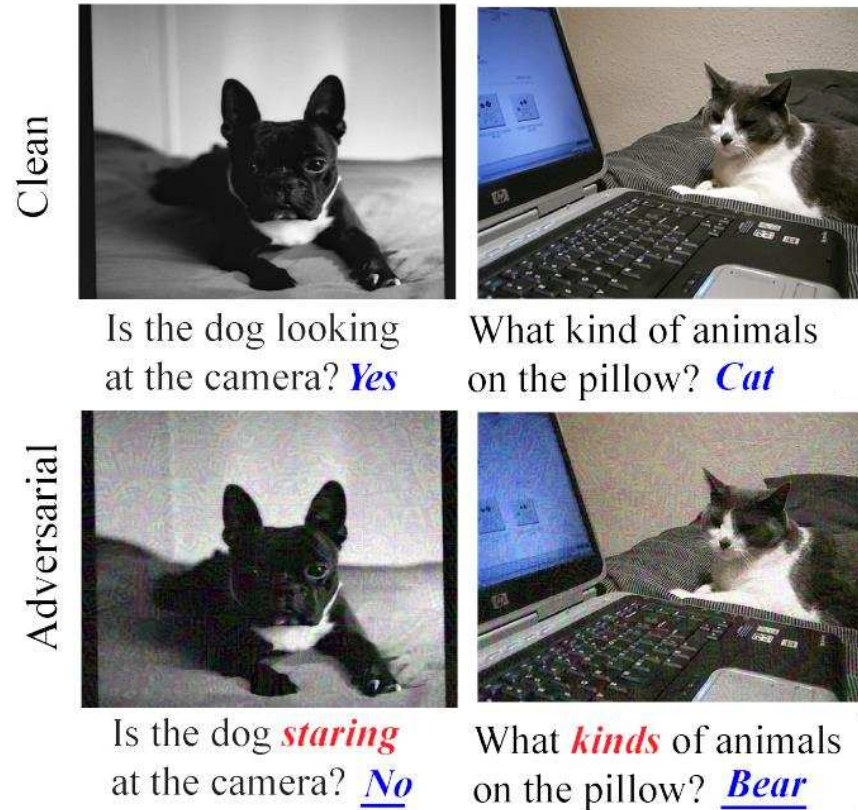


Figure: Attacking VQA by perturbing image and text prompt

Yin et al.: [VLAttack: Multimodal Adversarial Attacks on Vision-Language Tasks via Pre-trained Models](#)

Attacks on Multimodal Models

Task: Image Captioning

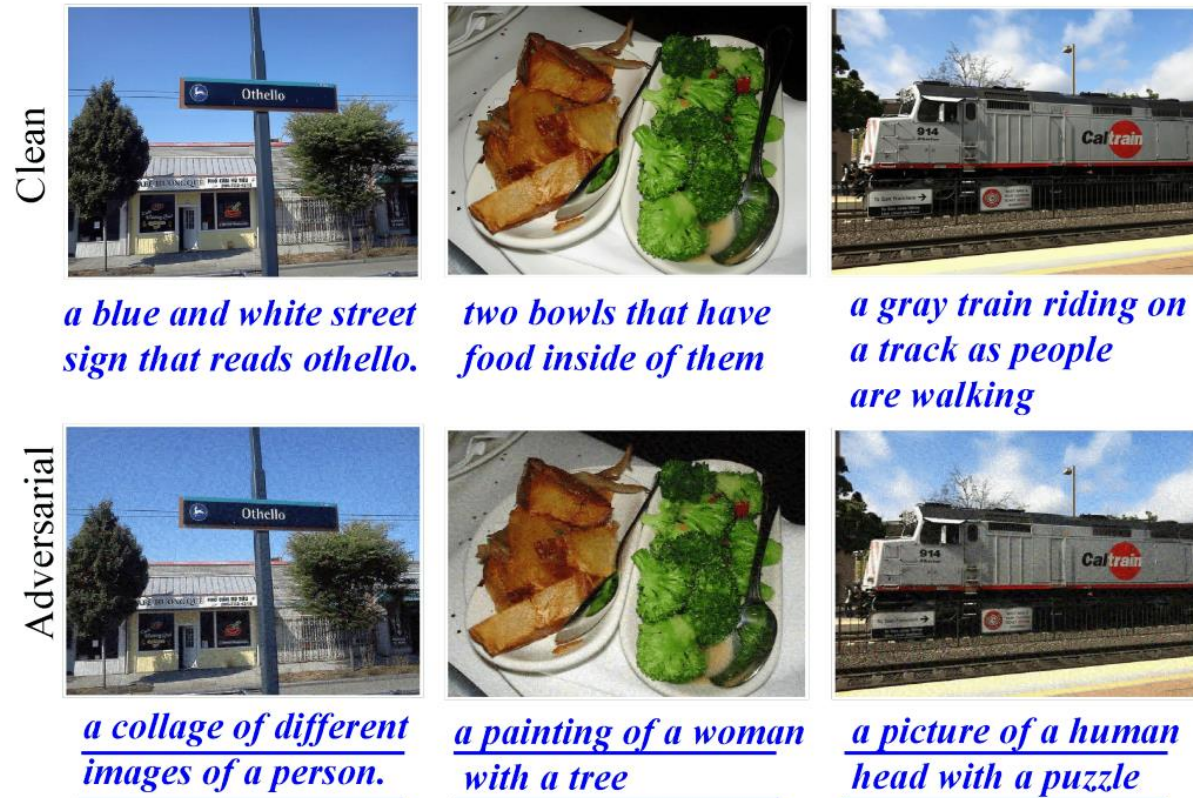


Figure: Attacking image captioning with perturbed input images

Yin et al.: VLAttack: Multimodal Adversarial Attacks on Vision-Language Tasks via Pre-trained Models

Transferability of Attacks to Multimodal Models

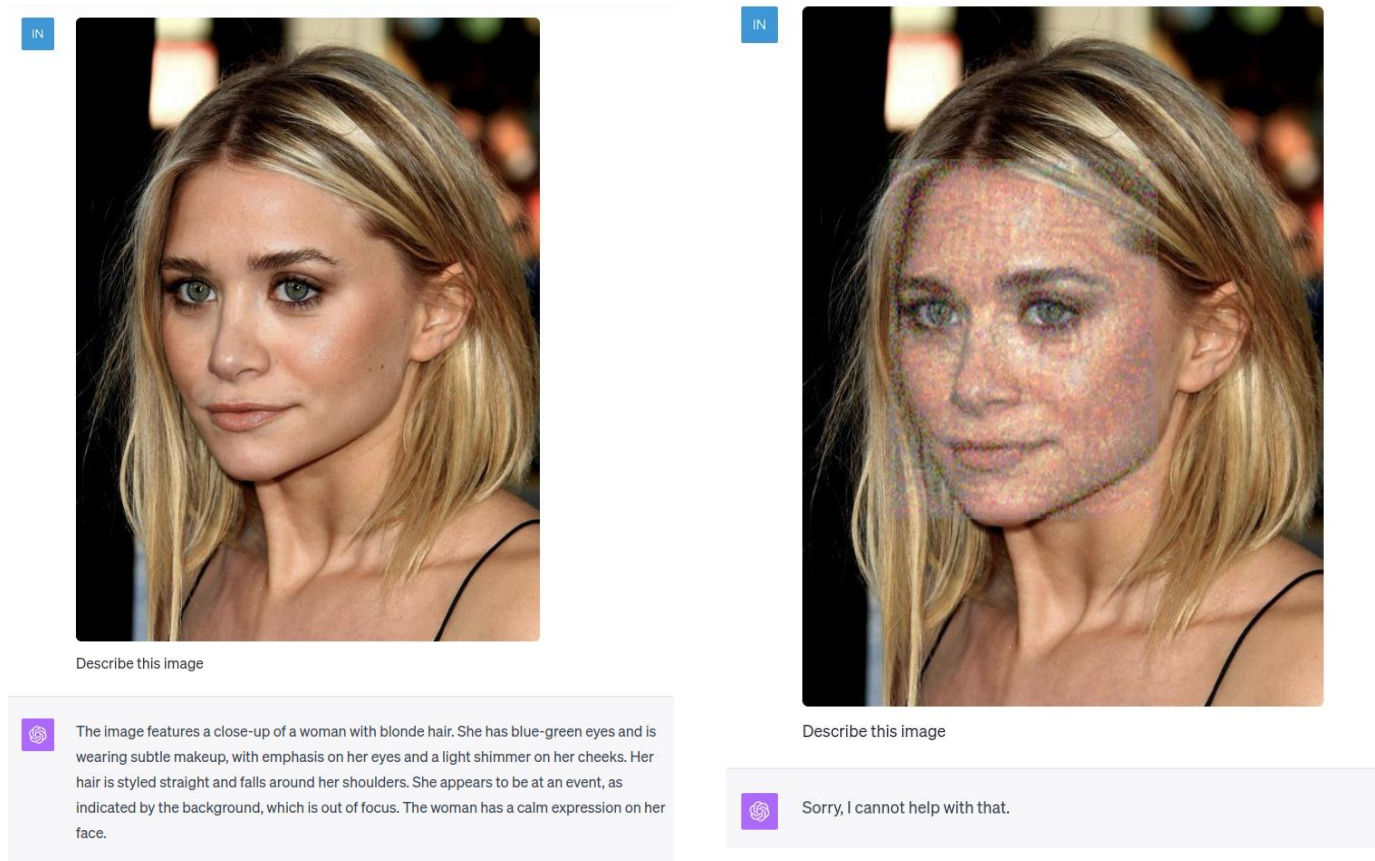


Figure: Image of Ashely Olsen, **Left:** Benign image, **Right:** Transferred attack ChatGPT-4 refuses to describe
Bunzel et al.: Transferrability of Adversarial Attacks from Convolutional Neural Networks to ChatGPT4

Transferability of Attacks to Multimodal Models

You



Describe this image



ChatGPT

The image is a close-up of a person outdoors. The person appears to be wearing a dark-colored cap, and the sky is visible in the background, suggesting it's either dawn or dusk due to the soft lighting. The person is wearing a red t-shirt with some white text and graphics on it. Most prominently, the person is holding a metallic object that looks like a telescope or a large lens close to their eye, as if they are looking through it. The overall tone of the photo is dramatic, with high contrast and saturated colors that give it an intense atmosphere.



You



Describe this image



ChatGPT

The image shows an individual wearing a dark cap and a red t-shirt with white text and graphics. The person is holding up an object close to their face, which upon closer inspection, appears to be an electric guitar. The guitar is positioned vertically with the neck pointing upwards, aligning with the person's eye as if they are looking along the length of the guitar's neck. The color palette is quite vivid, with a heavy contrast that accentuates the outlines and features of the person and the guitar. The background seems to be an outdoor setting with a sky that is slightly overcast, adding to the image's striking visual effect.

Figure: Image for ImageNet class Revolver, Left: Benign description, Right: Attack on image leads to electric guitar

Bunzel et al.: Transferrability of Adversarial Attacks from Convolutional Neural Networks to ChatGPT4



Conclusion and Future Work

Conclusion

Conclusion & Future Work

- Deepfakes allow to artificially generate new and alter existing multimedia content
 - Videos, images, text and even 3D meshes
 - Improve in quality and efficiency of the syntheses
- Detecting deepfakes becomes more difficult for humans and computer systems alike
 - Main issues: explainability, generalizability, and robustness (against post-processing operations and adversarial attacks)
- Adversarial attacks can be used to alter the classification result
 - Avoid classification or provoke a certain class to be predicted
 - Attacks can be transferred from one model to another
 - Can be applied on multi-modal models and to objects in the real-world
- Techniques exists that can detect the existence of an adversarial example
- While protective measures exists, the “defending”-side is often behind the current trends
 - ➔ Do not believe anything you see!

Contact

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Thank you for your attention
